Probability Theory & Mathematical Statistics 概率论和数理统计

Probability Space

1.1 Sample Space and Events

Sample space Ω is the set of all possible outcomes of an experiment.

An event is a subset $A \subseteq \Omega$.

Def σ-algebra

A family $\Sigma \subseteq 2^{\Omega}$ is a $\boldsymbol{\sigma}$**-algebra** if

$$\begin{aligned} &\emptyset \in \Sigma \\ &A \in \Sigma \Rightarrow A^c \in \Sigma \\ &A_1, A_2, \ldots \in \Sigma \Rightarrow \bigcup_i A_i \in \Sigma, \text{countably many} \end{aligned} \tag{1.1}$$

For a discrete probability space, a **probability mass function**,*pmf*, is a map

$$p : \Omega \to [0, 1], \quad \sum_{\omega \in \Omega} p(\omega) = 1 \tag{1.2}$$

and the **probability of an event** is

$$\Pr(A) = \sum_{\omega \in A} p(\omega) \tag{1.3}$$

1.2 Probability Measure

Def probability measure

A probability measure is a map $\Pr : \Sigma \to [0, 1]$ such that

$$\Pr(\Omega) = 1 \tag{2.1}$$

and for pairwise disjoint events $A_1, A_2, \ldots \in \Sigma$,

$$\Pr\left(\bigcup_i A_i\right) = \sum_i \Pr(A_i) \tag{2.2}$$

The triple $(\mathbf{\Omega}, \mathbf{\Sigma}, \mathbf{Pr})$ is called a **probability space**.

Classical probability on a *finite uniform sample space* satisfies

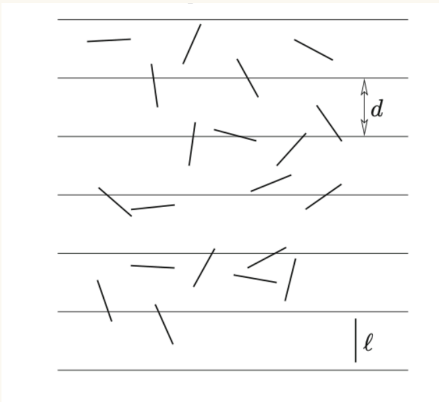
$$\Pr(A) = \frac{|A|}{|\Omega|} \tag{2.3}$$

and **geometric probability** satisfies

$$\Pr(A) = \frac{\mathrm{Vol}(A)}{\mathrm{Vol}(\Omega)} \tag{2.4}$$

Example

Buffon's needle problem.



If a needle of length $\ell < d$ is dropped onto ruled paper with gap d.

$$\begin{aligned} &x \in [0, \pi] : \text{angle between the needle \& line below} \\ &y \in \left[0, \frac{d}{2}\right] : \text{the distance between center of needle \& closest line} \end{aligned} \tag{2.5}$$

then with

$$\Omega = [0, \pi] \times \left[0, \frac{d}{2}\right] \tag{2.6}$$

and

$$A = \left\{ (x, y) \in \Omega : y \leq \frac{\ell}{2} \sin x \right\} \tag{2.7}$$

one gets

$$\Pr(A) = \frac{\mathrm{Vol}(A)}{\mathrm{Vol}(\Omega)} = \frac{2}{d\pi} \int_0^\pi \frac{\ell}{2} \sin x \, \mathrm{d}x = 2\frac{\ell}{d\pi} \tag{2.8}$$

1.3 Basic Properties

All of the following are consequences of the axioms:

$$\Pr(A^c) = 1 - \Pr(A) \tag{3.1}$$

$$\Pr(\emptyset) = 0 \tag{3.2}$$

$$\Pr(A \setminus B) = \Pr(A) - \Pr(A \cap B) \tag{3.3}$$

$$A \subseteq B \Rightarrow \Pr(A) \leq \Pr(B) \tag{3.4}$$

$$\Pr(A \cup B) = \Pr(A) + \Pr(B) - \Pr(A \cap B) \tag{3.5}$$

Note

自然数是偶数的概率是$\frac{1}{2}$: Not even wrong(无法定义一个σ-field). 但是$[0, 1]$中有理数概率是0是 well defined 的.

1.4 Union Bound

For events $A_1, ..., A_n$,

$$\Pr\left(\bigcup_{i=1}^{n} A_i\right) \leq \sum_{i=1}^{n} \Pr(A_i) \tag{4.1}$$

Example

If an algorithm has n types of errors and each type occurs with probability at most p, then

$$\Pr(\text{no error}) = 1 - \Pr\left(\bigcup_{i=1}^{n} A_i\right) \geq 1 - np \tag{4.2}$$

Ramsey Theory

If

$$\binom{n}{k} 2^{1-\binom{n}{2}} < 1 \tag{4.3}$$

then there exists a 2-coloring of K_n with no monochromatic K_k. Hence $R(k,k) > n$.

Proof

Color each edge of K_n independently red or blue with probability $\frac{1}{2}$. For each k-subset S of the vertices, let A_S be the event that the induced K_S is monochromatic. Then

$$\Pr(A_S) \overset{\text{two type of monochromatic}}{=} 2^{1-k\frac{k-1}{2}} \tag{4.4}$$

and by the union bound

$$\Pr\left(\bigcup_S A_S\right) \leq \frac{n!}{k!(n-k)!} 2^{1-k\frac{k-1}{2}} < 1 \tag{4.5}$$

so with positive probability there is no monochromatic K_k.

Example

Balls into bins. Throw n balls independently and uniformly into n bins. Every bin receives at most $O\left(\frac{\ln n}{\ln \ln n}\right)$ bins *w.h.p.*(with high probability), with probability $1 - O\left(\frac{1}{n}\right)$.

Proof. A: Some bin receives k balls (k to be fixed).

A_i: event that bin i receives at least k balls. Let

$$\Pr(A) = \Pr\left(\bigcup_{i=1}^{n} A_i\right) \sum_i \Pr(A_i) \leq \frac{1}{n} \Rightarrow \Pr(A^c) \geq 1 - \frac{1}{n} \tag{4.6}$$

For each $S \in \binom{[n]}{k} \in 2^{[n]}$, define $A_{i,S}$: bin-i receives the balls in S.

By union bound,

$$\begin{aligned} \Pr(A_i) &= \Pr\left(\bigcup_{S \in \binom{[n]}{k}} A_{i,S}\right) \leq \sum_{S \in \binom{[n]}{k}} \Pr(A_{i,S}) \\ &= \binom{n}{k} \frac{1}{n^k} \leq \left(\frac{en}{k}\right)^k \frac{1}{n^k} = \left(\frac{e}{k}\right)^k \leq \frac{1}{n^2}, \text{when } k = 3\frac{\ln n}{\ln \ln n} \end{aligned} \tag{4.7}$$

1.5 Inclusion-Exclusion

Principle of inclusion-exclusion

For events $A_1, \ldots, A_n$,

$$\begin{aligned}\Pr\left(\bigcup_{i=1}^{n} A_i\right) &= \sum_{i=1}^{n} \Pr(A_i) - \sum_{i<j} \Pr(A_i \cap A_j) + \sum_{i<j<k} \Pr(A_i \cap A_j \cap A_k) - \ldots \\ &= \sum_{\emptyset \neq S \subseteq [n]} (-1)^{|S|-1} \Pr\left(\bigcap_{i \in S} A_i\right)\end{aligned} \tag{5.1}$$

The Boole-Bonferroni inequalities

for every $A_1, A_2, \ldots, A_n \in \Sigma$, for any $k \geq 0$

$$\sum_{\substack{S \in \{1,2,\ldots,n\}, \\ 1 \leq |S| \leq 2k}} (-1)^{|S|-1} \Pr\left(\bigcap_{i \in S} A_i\right) \leq \Pr\left(\bigcup_{i=1}^{n} A_i\right) \leq \sum_{\substack{S \in \{1,2,\ldots,n\}, \\ 1 \leq |S| \leq 2k+1}} (-1)^{|S|-1} \Pr\left(\bigcap_{i \in S} A_i\right) \tag{5.2}$$

Example

Derangement. Let $\pi : [n] \to [n]$ be a uniformly random permutation and let A_i be the event $\pi(i) = i$. Then

$$\begin{aligned}\Pr(\text{pi has no fixed point}) &= 1 - \Pr\left(\bigcup_{i=1}^{n} A_i\right) \\ &= 1 - \sum_{k=1}^{n} \sum_{S \in \binom{\{1,2,\ldots,n\}}{k}} (-1)^{k-1} \Pr\left(\bigcap_{i \in S} A_i\right) \\ &= 1 + \sum_{k=1}^{n} \binom{n}{k} (-1)^k \frac{(n-k)!}{n!} \\ &= \sum_{k=0}^{n} (-1)^k \frac{1}{k!} \to \frac{1}{e}, n \to +\infty\end{aligned} \tag{5.3}$$

hence

$$\Pr(\text{pi has no fixed point}) \to \frac{1}{e} \tag{5.4}$$

1.6 Continuity and Null Events

If $A_1 \subseteq A_2 \subseteq \ldots$ and $A = \bigcup_{i=1}^{\infty} A_i$, then

$$
\begin{aligned}
\Pr(A) &= \Pr(A_1) \sum_{i=1}^{\infty} \Pr(A_{i+1} \setminus A_i) \\
&= \Pr(A_1) + \lim_{i=1}^{\infty} \sum_{i=1}^{n-1} (\Pr(A_i + 1) - \Pr(A_i)) \\
&= \lim_{i \to \infty} \Pr(A_i)
\end{aligned}
\tag{6.1}
$$

If $B_1 \supseteq B_2 \supseteq \ldots$ and $B = \bigcap_{i=1}^{\infty} B_i$, then

$$
\Pr(B) = \lim_{i \to \infty} \Pr(B_i) \tag{6.2}
$$

1.7 Null and Almost Surely Events

An event is called **null** if $\Pr(A) = 0$.
An event occurs **almost surely** if $\Pr(A) = 1$.
A **complete** probability space contains every subset of every null event.

Conditional Probability

2.1 Definition

For events A, B with $\Pr(B) > 0$, the **conditional probability** of A given B is

$$
\Pr(A \mid B) = \frac{\Pr(A \cap B)}{\Pr(B)} \tag{1.1}
$$

The map $A \Rightarrow \Pr(A \mid B)$ is itself a probability law on

$$
\Sigma_B = \{A \cap B : A \in \Sigma\} \tag{1.2}
$$

The triple is $\boldsymbol{(B, \Sigma_B \coloneqq \{A \cap B, A \in \Sigma\}, \Pr(\cdot \mid B))}$

2.2 Basic Examples

> **Example**
>
> Fair coin from a biased one. Repeatedly toss the biased coin until either HT or TH appears; output H in the first case and T in the second. Since
>
> $$\Pr(\mathrm{HT} \mid \{\mathrm{HT}, \mathrm{TH}\}) = \Pr(\mathrm{TH} \mid \{\mathrm{HT}, \mathrm{TH}\}) = \frac{1}{2} \tag{2.1}$$

this produces a fair coin from an unknown biased coin.

Example

Two-child problem. Under the uniform law on

$$\Omega = \{\mathrm{BB}, \mathrm{BG}, \mathrm{GB}, \mathrm{GG}\} \tag{2.2}$$

if we know that at least one child is a girl, then

$$\Pr(\mathrm{GG} \mid \{\mathrm{BG}, \mathrm{GB}, \mathrm{GG}\}) = \frac{1}{3} \tag{2.3}$$

2.3 Fundamental Laws

For events $A_1, \ldots, A_n$ with positive intermediate probabilities, the chain rule is

$$\Pr\left(\bigcap_{i=1}^{n} A_i\right) = \prod_{i=1}^{n} \Pr\left(A_i \mid \bigcap_{j<i} A_j\right) \tag{3.1}$$

If $B_1, \ldots, B_n$ is a *partition* of Ω, then **the law of total probability** is

$$\Pr(A) = \sum_{i=1}^{n} \Pr(A \mid B_i) \Pr(B_i) \tag{3.2}$$

Under the same partition, Bayes' law is

$$\Pr(B_i \mid A) = \frac{\Pr(B_i) \Pr(A \mid B_i)}{\sum_{j=1}^{n} \Pr(B_j) \Pr(A \mid B_j)} \tag{3.3}$$

2.4 Conditional Probability Examples

Birthday Paradox

For a uniform random map $f : [n] \to [m]$,

$$\Pr(f \text{ is 1-1}) = \left(\frac{m!}{(m-n)!} \frac{1}{m^n}\right) = \prod_{i=1}^{n} \left(1 - \frac{i-1}{m}\right) \tag{4.1}$$

and

$$\begin{aligned} \prod_{i=1}^{n} \left(1 - \frac{i-1}{m}\right) &\approx \prod_{i=1}^{n} \exp\left(\frac{i-1}{m}\right) \\ &\approx \exp\left(\sum_{i=1}^{n} \left(\frac{i-1}{m}\right)\right) \approx \exp\left(-\frac{n^2}{2m}\right) \end{aligned} \tag{4.2}$$

Apply it to **Balls into bins**, throwing n in m one-by-one.

$$\begin{aligned} \Pr(\text{Every ball is thrown into an empty bin}) = \varepsilon = \exp\left(-\frac{n^2}{2m}\right) \\ \Rightarrow \sqrt{2m \ln\left(\frac{1}{\varepsilon}\right)} \end{aligned} \tag{4.3}$$

Monty Hall Problem

Let A be the event that you win the car in the end, and let B be the event that your first choice is already the car.

If you do not switch, then

$$\Pr(A) = \Pr(B) = \frac{1}{3} \tag{4.4}$$

If you always switch, then by the law of total probability

$$\begin{aligned} \Pr(A) &= \Pr(A \mid B)\Pr(B) + \Pr(A \mid B^c)\Pr(B^c) \\ &= 0 \cdot \frac{1}{3} + 1 \cdot \frac{2}{3} = \frac{2}{3} \end{aligned} \tag{4.5}$$

so switching doubles the success probability.

Gambler's Ruin

Let A be the event that the gambler is ruined before reaching n, and let $\Pr_k$ denote the law started from capital k.

Conditioning on the first toss $B = \{\text{the first toss is HEAD}\}$,

$$\begin{aligned} \Pr_{k(A)} &= \Pr_{k(A \mid B)} \Pr(B) + \Pr_{k(A \mid B^c)} \Pr(B^c) \\ &= \frac{1}{2} \Pr_{k+1}(A) + \frac{1}{2} \Pr_{k-1}(A) \end{aligned} \tag{4.6}$$

for $0 < k < n$, with boundary conditions

$$\Pr_0(A) = 1, \quad \Pr_n(A) = 0 \tag{4.7}$$

Write $u_k = \Pr_k(A)$. Then

$$u_{k+1} - u_k = u_k - u_{k-1} \tag{4.8}$$

so (u_k) is arithmetic and therefore

$$u_k = \alpha + \beta k \tag{4.9}$$

Using $u_0 = 1$ and $u_n = 0$, we get $\alpha = 1$ and $\beta = -\frac{1}{n}$. Hence

$$\Pr_{k(A)} = u_k = 1 - \frac{k}{n} \tag{4.10}$$

Dominating False Positives

Let I be the event that the person is ill. We are given

$$\Pr(I) = 0.001, \quad \Pr(+ \mid I) = 0.95, \quad \Pr(+ \mid I^c) = 0.05 \tag{4.11}$$

By the law of total probability,

$$\begin{aligned} \Pr(+) &= \Pr(+ \mid I)\Pr(I) + \Pr(+ \mid I^c)\Pr(I^c) \\ &= 0.95 \cdot 0.001 + 0.05 \cdot 0.999 \end{aligned} \tag{4.12}$$

Therefore, by Bayes' law,

$$\begin{aligned} \Pr(I \mid +) &= \frac{\Pr(+ \mid I)\Pr(I)}{\Pr(+)} \\ &= \frac{0.95 \cdot 0.001}{0.95 \cdot 0.001 + 0.05 \cdot 0.999} \\ &\approx 1.87\% \end{aligned} \tag{4.13}$$

So even a positive test still leaves the disease unlikely, because false positives dominate the prior.

Simpson's Paradox

In the drug-trial example from the PPT:

$$\begin{aligned} \Pr(\text{success} \mid \text{Drug I, women}) &= \frac{200}{2000} = \frac{1}{10} > \frac{10}{200} \\ &= \frac{1}{20} = \Pr(\text{success} \mid \text{Drug II, women}) \end{aligned} \tag{4.14}$$

$$\begin{aligned} \Pr(\text{success} \mid \text{Drug I, men}) &= \frac{19}{20} \\ &> \frac{1000}{2000} = \frac{1}{2} = \Pr(\text{success} \mid \text{Drug II, men}) \end{aligned} \tag{4.15}$$

But after aggregation,

$$\Pr(\text{success} \mid \text{Drug I}) = \frac{219}{2020} < \frac{1010}{2200} = \Pr(\text{success} \mid \text{Drug II}) \tag{4.16}$$

Abstractly, it is possible that for every block C_i of a partition,

$$\Pr(A \mid B \cap C_i) > \Pr(A \mid B^c \cap C_i) \tag{4.17}$$

while overall one still has

$$\Pr(A \mid B) < \Pr(A \mid B^c) \tag{4.18}$$

Independence

3.1 Independence of Events

Two events A and B are **independent** if

$$\Pr(A \cap B) = \Pr(A)\Pr(B) \tag{1.1}$$

If $\Pr(B) > 0$, this is equivalent to

$$\Pr(A \mid B) = \Pr(A) \tag{1.2}$$

and also equivalent to

$$\Pr(A \cap B^c) = \Pr(A)\Pr(B^c) \tag{1.3}$$

For a family $\{A_i : i \in I\}$, **mutual independence** means that for every finite $J \subseteq I$,

$$\Pr\left(\bigcap_{i \in J} A_i\right) = \prod_{i \in J} \Pr(A_i) \tag{1.4}$$

An event A is independent of a family $\{B_i : i \in I\}$ if for all disjoint finite $J_+, J_- \subseteq I$,

$$\Pr\left(A \mid \bigcap_{i \in J_+} B_i \cap \bigcap_{i \in J_-} B_i^c\right) = \Pr(A) \tag{1.5}$$

3.2 Product Probability Space

Given discrete probability spaces $(\Omega_1, p_1), \ldots, (\Omega_n, p_n)$, the product space is

$$\Omega = \Omega_1 \times \ldots \times \Omega_n \tag{2.1}$$

with product pmf

$$p(\omega_1, \ldots, \omega_n) = p_1(\omega_1) \ldots p_{n(\omega_n)} \tag{2.2}$$

For measurable rectangles one has

$$\Pr(A_1 \times \ldots \times A_n) = \Pr_1(A_1) \ldots \Pr_n(A_n) \tag{2.3}$$

and for general probability spaces one similarly defines the product σ-algebra and the corresponding product measure.

3.3 Limited Independence

Pairwise independence only requires

$$\Pr(A_i \cap A_j) = \Pr(A_i)\Pr(A_j) \quad \text{for all distinct } i, j \tag{3.1}$$

Mutual independence implies pairwise independence, but the converse fails.

Pairwise But Not Mutually Independent

Toss three fair coins. Let

$$D = \{X_1 \neq X_2\}, \quad E = \{X_2 \neq X_3\}, \quad F = \{X_3 \neq X_1\} \tag{3.2}$$

Each event has probability $\frac{1}{2}$, and

$$\Pr(D \cap E) = \Pr(D \cap F) = \Pr(E \cap F) = \frac{1}{4} = \left(\frac{1}{2}\right)\left(\frac{1}{2}\right) \tag{3.3}$$

so they are pairwise independent.

But

$$D \cap E \cap F = \emptyset \tag{3.4}$$

because three bits cannot all be pairwise different. Hence

$$\Pr(D \cap E \cap F) = 0 \neq \frac{1}{8} = \Pr(D)\Pr(E)\Pr(F) \tag{3.5}$$

so they are not mutually independent.

Triple Product Without Pairwise Independence

The condition

$$\Pr(A \cap B \cap C) = \Pr(A)\Pr(B)\Pr(C) \tag{3.6}$$

by itself is much weaker than mutual independence.

For example, on $\Omega = \{000, 010, 101, 111\}$ with probabilities

$$\Pr(000) = \frac{1}{8}, \quad \Pr(010) = \frac{3}{8}, \quad \Pr(101) = \frac{3}{8}, \quad \Pr(111) = \frac{1}{8} \tag{3.7}$$

let A, B, C be the events that the first, second, third bit is 1. Then

$$\Pr(A) = \Pr(B) = \Pr(C) = \frac{1}{2} \tag{3.8}$$

and

$$\Pr(A \cap B \cap C) = \Pr(111) = \frac{1}{8} = \Pr(A)\Pr(B)\Pr(C) \tag{3.9}$$

but

$$\Pr(A \cap B) = \frac{1}{8} \neq \frac{1}{4} = \Pr(A)\Pr(B) \tag{3.10}$$

so no pair is independent.

3.4 Independent Repetition

One-Sided Error Reduction

Suppose a randomized algorithm $\mathcal{A}$ has one-sided error:

$$f(x) = 1 \Rightarrow \mathcal{A}(x) = 1, \quad f(x) = 0 \Rightarrow \Pr(\mathcal{A}(x) = 0) \geq p \tag{4.1}$$

Run the algorithm independently n times and return the conjunction of the outputs. On a no-instance, an error occurs only if every run is wrong, so by independence

$$\Pr(\text{wrong}) \leq (1 - p)^n \tag{4.2}$$

Therefore it is enough to take

$$n \geq \frac{\ln\left(\frac{1}{\varepsilon}\right)}{p} \tag{4.3}$$

to reduce the error to at most ε.

Binomial Probability

For n independent Bernoulli trials with success probability p, the probability of exactly k successes is

$$
\begin{aligned}
p(k) & \\
&= \sum_{S \in \binom{[n]}{k}} \Pr(\text{trials in } S \text{ succeed}) \Pr(\text{trials in } ([n] \setminus S) \text{ fail}) \\
&= \sum_{S \in \binom{[n]}{k}} p^k (1-p)^{n-k} \qquad (4.4) \\
&= \frac{n!}{k!(n-k)!} p^k (1-p)^{n-k}
\end{aligned}
$$

Hence $p(k)$ is a pmf on $\{0, 1, \ldots, n\}$, and

$$\sum_{k=0}^{n} p(k) = 1 \tag{4.5}$$

Fair Voting

Let X be the number of heads in n independent tosses of a fair coin. Then

$$
\begin{aligned}
\Pr(|2X - n| \geq t) &= \Pr\left(X \leq \frac{n-t}{2}\right) + \Pr\left(X \geq \frac{n+t}{2}\right) \\
&= 2^{1-n} \sum_{k \leq \frac{n-t}{2}} \frac{n!}{k!(n-k)!}
\end{aligned} \tag{4.6}
$$

Using the entropy bound on the volume of a Hamming ball,

$$\Pr(|2X - n| \geq t) \leq 2^{1-n+nH\left(\frac{1}{2} - \frac{t}{2n}\right)} \approx 2 \exp\left(-\frac{t^2}{n}\right) \tag{4.7}$$

Thus to make this probability at most 0.05, it is enough to require

$$t \geq 2\sqrt{n} \tag{4.8}$$

Two-Sided Error Reduction

Suppose

$$\Pr(\mathcal{A}(x) = f(x)) \geq \frac{1}{2} + p \tag{4.9}$$

for every input x. Run the algorithm independently n times and return the majority answer. Then an error means that fewer than half of the runs are correct, so

$$\Pr(\text{majority is wrong}) \leq \sum_{k < \frac{n}{2}} \frac{n!}{k!(n-k)!} \left(\frac{1}{2} + p\right)^k \left(\frac{1}{2} - p\right)^{n-k} \tag{4.10}$$

and the **concentration inequalities** gives the upper bound

$$\Pr(\text{majority is wrong}) \leq \exp(-p^2 n) \tag{4.11}$$

Hence it is enough to take

$$n \geq \frac{\ln\left(\frac{1}{\varepsilon}\right)}{p^2} \tag{4.12}$$

3.5 Network Reliability and Conditional Independence

Serial-Parallel(串并联) Network Reliability

If each edge is present independently with probabilities $p_1, ..., p_5$, then

$$P_{AB} = 1 - (1 - p_1)(1 - p_2)(1 - p_3) \tag{5.1}$$

because A and B are connected iff at least one of the three parallel edges is present.

Then

$$P_{AC} = P_{AB} p_5, \quad P_{DE} = p_4 \tag{5.2}$$

and the two branches from s to t work independently, so

$$P_{st} = 1 - (1 - P_{AC})(1 - P_{DE}) \tag{5.3}$$

that is,

$$P_{st} = 1 - (1 - P_{AB} p_5)(1 - p_4) \tag{5.4}$$

3.6 Conditional independence

For events A, B, C with $\Pr(C) > 0$, we say that A and B are conditionally independent given C if

$$\Pr(A \cap B \mid C) = \Pr(A \mid C) \Pr(B \mid C) \tag{6.1}$$

If also $\Pr(B \cap C) > 0$, this is equivalent to

$$\Pr(A \mid B \cap C) = \Pr(A \mid C) \tag{6.2}$$

Conditional Independence Example

Toss two fair coins. Let

$$A = \{X_1 = H\}, \quad B = \{X_2 = H\}, \quad C = \{X_1 \neq X_2\} \tag{6.3}$$

Then A and B are independent, because

$$\Pr(A \cap B) = \frac{1}{4} = \Pr(A)\Pr(B) \tag{6.4}$$

But conditioned on C, they are no longer independent:

$$\Pr(A \cap B \mid C) = 0 \tag{6.5}$$

while

$$\Pr(A \mid C)\Pr(B \mid C) = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4} \tag{6.6}$$

So **independence does not imply conditional independence**.

Random Variables

4.1 Random Variable and Distribution

A random variable on a probability space $(\Omega, \Sigma, \Pr)$ is a function

$$X : \Omega \to \mathbb{R} \tag{1.1}$$

satisfying

$$\forall x \in \mathbb{R}, \{\omega \in \Omega : X(\omega) \leq x\} \in \Sigma \tag{1.2}$$

that is, X is Σ-measurable.

For discrete random variables $X : \Omega \to \mathbb{Z}$, this implies that every event of the form

$$\{X \in S\}, \quad S \subseteq \mathbb{Z} \tag{1.3}$$

is measurable.

Example

Roll a die and let X be the face value. Let $Y \in \{0, 1\}$ indicate whether the outcome is odd. Then X and Y are both random variables on the same sample space, but they carry different amounts of information.

Def CDF

The **cumulative distribution function** of X is

$$F_X(x) = \Pr(X \leq x) \tag{1.4}$$

Two random variables X and Y are **identically distributed** if

$$F_X = F_Y \tag{1.5}$$

Basic properties of the CDF:

$$\begin{aligned} & x \leq y \Rightarrow F_X(x) \leq F_{X(y)} \\ & \lim_{x \to -\infty} F_X(x) = 0 \\ & \lim_{x \to +\infty} F_X(x) = 1 \end{aligned} \tag{1.6}$$

4.2 Discrete and Continuous Random Variables

Def **discrete random variable**

X is **discrete** if $X(\Omega)$ is countable. Its **probability mass function** is

$$p_X(x) = \Pr(X = x) \tag{2.1}$$

and

$$F_{X(y)} = \sum_{x \leq y} p_X(x) \tag{2.2}$$

Def **continuous random variable**

X is **continuous** if there exists an integrable density f_X such that

$$F_{X(y)} = \Pr(X \leq y) = \int_{-\infty}^{y} f_X(x) \, \mathrm{d}x \tag{2.3}$$

Note

There exist random variables that are neither discrete nor continuous.

4.3 Independence and Random Vector

Two random variables X and Y are **independent** if the events $X \leq x$ and $Y \leq y$ are independent for every $x, y \in \mathbb{R}$.

For **discrete random variables**, this is equivalent to

$$\Pr(X = x \cap Y = y) = \Pr(X = x) \Pr(Y = y) \tag{3.1}$$

for all x, y.

For random variables $X_1, \ldots, X_n$, **mutual independence** means

$$\Pr(X_1 = x_1 \cap \ldots \cap X_n = x_n) = p_{X_1}(x_1) \ldots p_{X_n}(x_n) \tag{3.2}$$

for every choice of values.

Def **random vector**

A random vector is an n-tuple

$$X = (X_1, \ldots, X_n) \tag{3.3}$$

where each X_i is a random variable. Its **joint CDF** is

$$F_{X(x_1, \ldots, x_n)} = \Pr(X_1 \leq x_1 \cap \ldots \cap X_n \leq x_n) \tag{3.4}$$

For a discrete random vector, the joint pmf is

$$p_{X(x_1, \ldots, x_n)} = \Pr(X_1 = x_1 \cap \ldots \cap X_n = x_n) \tag{3.5}$$

and the marginal distribution is obtained by summing out the remaining coordinates.

Example of a pairwise independence

The construction of $2^n - 1$ pairwise independent random bits out of mutually independent random bits by XOR.

That is, $n \approx 2^k$, we want to generate $I_i \in \{0, 1\}, i \in [n]$, that is

$$\Pr(I_i = 1) = \frac{1}{2} \tag{3.6}$$

and I_i is pairwise independent. There's a $\Theta(n)$ algorithm:
k-bits random bits generate one-by-one: $X := \underbrace{000\ldots00}_{k \text{ bits}} \to 111\ldots11$ $B_{i,j}$ is the j-th bit of i encoding into binary.

$$\Pr(I_i = 1) := \bigoplus_{B_{i,j}=1} B(X, j) \tag{3.7}$$

Discrete Distributions

5.1 Bernoulli Trial

Def Bernoulli distribution

A Bernoulli random variable takes values in $\{0, 1\}$ and satisfies

$$p_{X(1)} = p, \quad p_{X(0)} = 1 - p \tag{1.1}$$

where $p \in [0, 1]$.

For an event A, its indicator random variable is

$$\begin{aligned} I(A) &= 1, \text{if } A \text{ occurs} \\ I(A) &= 0, \text{otherwise} \end{aligned} \tag{1.2}$$

which is a Bernoulli random variable with parameter $\Pr(A)$.

5.2 Binomial Distribution

Def binomial distribution

If X is the number of successes in n i.i.d. Bernoulli trials with parameter p, then

$$p_{X(k)} = \Pr(X = k) = \binom{n}{k} p^k (1-p)^{n-k}, \quad k = 0, 1, \ldots, n \tag{2.1}$$

We write $X \sim \mathrm{Bin}(n, p)$.

Note

One may write

$$X = X_1 + \ldots + X_n \tag{2.2}$$

where $X_1, \ldots, X_n$ are i.i.d. Bernoulli random variables.

5.3 Geometric Distribution

Def geometric distribution

If X is the number of Bernoulli trials needed to get the first success, then

$$p_{X(k)} = \Pr(X = k) = (1-p)^{k-1}p, \quad k = 1, 2, \ldots \tag{3.1}$$

We write $X \sim \mathrm{Geo}(p)$.

Memoryless Property

For $X \sim \mathrm{Geo}(p)$, for every $k \geq 1$ and $n \geq 0$,

$$\Pr(X = k + n \mid X > n) = \Pr(X = k) \tag{3.2}$$

Proof.

$$\begin{aligned} \Pr(X = k + n \mid X > n) &= \frac{\Pr(X = k + n)}{\Pr(X > n)} \\ &= \frac{(1-p)^{n+k-1}p}{(1-p)^n} \\ &= (1-p)^{k-1}p = \Pr(X = k) \end{aligned} \tag{3.3}$$

Note

The geometric distribution is the only discrete memoryless distribution on $\{1, 2, \ldots\}$.

5.4 Sum of Independent Random Variables

If discrete random variables X and Y are independent, then

$$p_{X+Y}(z) = \sum_x \Pr(X = x \cap Y = z - x) = \sum_x p_X(x) p_{Y(z-x)} \tag{4.1}$$

This is the **convolution** of the two mass functions:

$$p_{X+Y} = p_X * p_Y \tag{4.2}$$

Proof

For i.i.d. Bernoulli random variables $X_1, \ldots, X_n$,

$$\begin{aligned} p_{X_1+\ldots+X_n}(k) &= p \cdot p_{X_1+\ldots+X_{n-1}}(k-1) + (1-p) \cdot p_{X_1+\ldots+X_{n-1}}(k) \\ &= \binom{n-1}{k-1} p^k (1-p)^{n-k} + \binom{n-1}{k} p^k (1-p)^{n-k} \\ &= \binom{n}{k} p^k (1-p)^{n-k} \end{aligned} \tag{4.3}$$

5.5 Negative Binomial Distribution

Def negative binomial distribution

Let X be the number of failures before the r-th success in i.i.d. Bernoulli trials. Then

$$p_{X(k)} = \Pr(X = k) = \binom{k + r - 1}{k}(1 - p)^k p^r, \quad k = 0, 1, 2, \ldots \tag{5.1}$$

Moreover,

$$X = (X_1 - 1) + \ldots + (X_r - 1) \tag{5.2}$$

where $X_1, \ldots, X_r$ are i.i.d. geometric random variables with parameter p.

5.6 Hypergeometric Distribution

Def hypergeometric distribution

Let X be the number of successes in n draws without replacement from a population of N objects containing exactly M successes. Then

$$p_{X(k)} = \Pr(X = k) = \frac{\binom{M}{k}\binom{N-M}{n-k}}{\binom{N}{n}} \tag{6.1}$$

5.7 Multinomial Distribution

Def multinomial distribution

Throw n balls independently into m bins, where bin i is chosen with probability p_i and

$$p_1 + \ldots + p_m = 1 \tag{7.1}$$

If X_i is the number of balls in bin i, then

$$\Pr(X_1 = k_1 \cap \ldots \cap X_m = k_m) = \frac{n!}{k_{1!} \ldots k_{m!}} p_1^{k_1} \ldots p_m^{k_m} \tag{7.2}$$

whenever $k_1 + \ldots + k_m = n$.

Each marginal satisfies

$$X_i \sim \operatorname{Bin}(n, p_i) \tag{7.3}$$

5.8 Poisson Distribution

Def **Poisson distribution**

A Poisson random variable with parameter $\lambda > 0$ satisfies

$$p_{X(k)} = \Pr(X = k) = \exp(-\lambda)\frac{\lambda^k}{k!}, \quad k = 0, 1, 2, \ldots \tag{8.1}$$

Poisson as Binomial Limit

If $n \to \infty$ and $np = \lambda$, then

$$\operatorname{Bin}\left(n, \frac{\lambda}{n}\right)(k) = \binom{n}{k}\left(\frac{\lambda}{n}\right)^k\left(1 - \frac{\lambda}{n}\right)^{n-k} \approx \exp(-\lambda)\frac{\lambda^k}{k!} \tag{8.2}$$

so the Poisson law is the idealized binomial distribution in the sparse regime.

Sum of Poisson Variables

If $X \sim \operatorname{Pois}(\lambda_1)$ and $Y \sim \operatorname{Pois}(\lambda_2)$ are independent, then

$$\begin{aligned} p_{X+Y}(k) &= \sum_{i=0}^{k} p_{X(i)} p_{Y(k-i)} \\ &= \sum_{i=0}^{k} \exp(-\lambda_1)\frac{\lambda_1^i}{i!}\exp(-\lambda_2)\frac{\lambda_2^{k-i}}{(k-i)!} \\ &= \frac{\exp(-(\lambda_1 + \lambda_2))}{k!}\sum_{i=0}^{k}\binom{k}{i}\lambda_1^i \lambda_2^{k-i} \\ &= \exp(-(\lambda_1 + \lambda_2))\frac{(\lambda_1 + \lambda_2)^k}{k!} \end{aligned} \tag{8.3}$$

Hence

$$X + Y \sim \operatorname{Pois}(\lambda_1 + \lambda_2) \tag{8.4}$$

Poisson Approximation for Balls into Bins

If $(X_1, \ldots, X_m)$ is multinomial with parameters $(m, n, p_1, \ldots, p_m)$ and $Y_i \sim \operatorname{Pois}(\lambda_i)$ are independent with $\lambda_i = np_i$, then $(X_1, \ldots, X_m)$ has the same distribution as $(Y_1, \ldots, Y_m)$ conditioned on $\sum_i Y_i = n$

Proof.

$$\sum_i Y_i \sim \mathrm{Pois}\left(\sum_i \lambda_i\right) = \mathrm{Pois}(n) \tag{8.5}$$

$$\begin{aligned} &\Pr\left(Y_i = y_i \mid \sum_i Y_i = n\right) \\ &= \exp\left(\sum_i \lambda_i\right) \prod_i \frac{\lambda_i^{y_i}}{y_i!} \cdot \left(e^n \frac{n^n}{n!}\right)^{-1} \\ &= \frac{n!}{\prod_i y_i!} \prod_i p_i^k \\ &= \Pr((X_1, X_2, ..., X_n) = (y_1, y_2, ..., y_n)) \end{aligned} \tag{8.6}$$

5.9 Random Objects

Example

In the balls-into-bins model, $(X_1, ..., X_m)$ follows a multinomial distribution with parameters $\left(m, n, \left(\frac{1}{m}, ..., \frac{1}{m}\right)\right)$.

Example

In the Erdős-Rényi random graph $G(n, p)$, the number of edges is

$$\mathrm{Bin}\left(\binom{n}{2}, p\right) \tag{9.1}$$

Example

In the Galton-Watson branching process,

$$X_{n+1} = \sum_{j=1}^{X_n} \xi_j^n \tag{9.2}$$

where the offspring variables are i.i.d. nonnegative integer-valued random variables.

5.10 Expectation

5.11 Definition

For a discrete random variable X, the **expectation** of X is

$$\mathbb{E}[X] = \sum_x x p_X(x) \tag{11.1}$$

assuming $\mathbb{E}[X] < \infty$.

The St. Petersburg paradox

$$p_X(2^k) = 2^{-k}, k = 1, 2, 3... \Rightarrow \mathbb{E}[X] = \infty \tag{11.2}$$

5.12 Expectation of Indicators

For a Bernoulli random variable with parameter p,

$$\mathbb{E}[X] = p \tag{12.1}$$

In particular, for an indicator variable,

$$\mathbb{E}[I(A)] = \Pr(A) \tag{12.2}$$

5.13 Expectation of Poisson

For $X \sim \text{Pois}(\lambda)$,

$$\begin{aligned} \mathbb{E}[X] &= \sum_{k \geq 0} k \exp(-\lambda) \frac{\lambda^k}{k!} \\ &= \lambda \sum_{k \geq 1} \exp(-\lambda) \frac{\lambda^{k-1}}{(k-1)!} \\ &= \lambda \end{aligned} \tag{13.1}$$

5.14 Change of Variables

Law of The Unconscious Statistician, LOTUS

For a function $f : \mathbb{R} \to \mathbb{R}$ and a discrete random variable X,

$$\mathbb{E}[f(X)] = \sum_x f(x) p_X(x) \tag{14.1}$$

More generally, for a discrete random vector $X = (X_1, ..., X_n)$,

$$\mathbb{E}[f(X_1, \ldots, X_n)] = \sum_{x_1,\ldots,x_n} f(x_1, \ldots, x_n) p_{X(x_1,\ldots,x_n)} \tag{14.2}$$

5.15 Linearity of Expectation

For random variables X, Y and scalars a, b,

$$\mathbb{E}[aX + b] = a\mathbb{E}[X] + b \tag{15.1}$$

$$\mathbb{E}[X + Y] = \mathbb{E}[X] + \mathbb{E}[Y] \tag{15.2}$$

Note

This holds whether or not X and Y are independent.

For any affine(or linear) function f,

$$\mathbb{E}[f(X_1, \ldots, X_n)] = f(\mathbb{E}[X_1], \ldots, \mathbb{E}[X_n]) \tag{15.3}$$

5.16 Expectations of Basic Distributions

Binomial Expectation

If $X \sim \mathrm{Bin}(n, p)$ and

$$X = X_1 + \ldots + X_n \tag{16.1}$$

with i.i.d. Bernoulli variables X_i, then by linearity

$$\mathbb{E}[X] = \mathbb{E}[X_1] + \ldots + \mathbb{E}[X_n] = np \tag{16.2}$$

Geometric Expectation

For $X \sim \mathrm{Geo}(p)$, write

$$X = \sum_{k\geq 1} I_k \tag{16.3}$$

where I_k indicates that the first $k - 1$ trials all fail. Then

$$\mathbb{E}[X] = \sum_{k\geq 1} \mathbb{E}[I_k] = \sum_{k\geq 1} (1 - p)^{k-1} = \frac{1}{p} \tag{16.4}$$

Negative Binomial Expectation

For negative binomial expectation

$$X = \sum_{k \geq 1} k \binom{k + r - 1}{k} (1 - p)^k p^r \quad (16.5)$$

$$X = (X_1 - 1) + \ldots + (X_r - 1), X_i \sim \mathrm{Geo}(p) \quad (16.6)$$

with i.i.d. $X_i \sim \mathrm{Geo}(p)$, then

$$\mathbb{E}[X] = r\mathbb{E}[X_1] - r = r\frac{1 - p}{p} \quad (16.7)$$

Hypergeometric Expectation

Let X count the red balls drawn. If X_i indicates whether the i-th red ball is drawn, then

$$X = X_1 + \ldots + X_M \quad (16.8)$$

and each red ball is drawn with probability $\frac{n}{N}$. Hence

$$\mathbb{E}[X] = \mathbb{E}[X_1] + \ldots + \mathbb{E}[X_M] = n\frac{M}{N} \quad (16.9)$$

5.17 Pattern Matching and Coupon Collector

Pattern Matching

Let $s = (s_1, \ldots, s_n) \in Q^n$ be a uniform random string over an alphabet Q with $|Q| = q$. For a fixed pattern $\pi \in Q^k$, let X be the number of appearances of π as a substring.

If I_i indicates that

$$\pi = (s_i, s_{i+1}, \ldots, s_{i+k-1}) \quad (17.1)$$

then

$$X = \sum_{i=1}^{n-k+1} I_i \quad (17.2)$$

and therefore

$$\mathbb{E}[X] = \sum_{i=1}^{n-k+1} \mathbb{E}[I_i] = (n - k + 1)q^{-k} \quad (17.3)$$

Coupon Collector

Let X be the number of boxes opened until all n coupon types are collected. Decompose

$$X = X_1 + \ldots + X_n \tag{17.4}$$

where X_i is the waiting time while there are exactly $i - 1$ collected types.

Then X_i is geometric with parameter

$$p_i = \frac{n - i + 1}{n} \tag{17.5}$$

so

$$\mathbb{E}[X] = \sum_{i=1}^{n} \mathbb{E}[X_i] = \sum_{i=1}^{n} \frac{n}{n - i + 1} = n \sum_{i=1}^{n} \frac{1}{i} \approx n \ln n \tag{17.6}$$

5.18 Double Counting Identity

For every nonnegative integer-valued random variable X, that X takes the value in $\{0, 1, 2, 3\ldots\}$,

$$\mathbb{E}[X] = \sum_{k \geq 0} \Pr(X > k) \tag{18.1}$$

Proof

Method 1:

Let I_k indicate the event $X > k$. Then

$$X = \sum_{k \geq 0} I_k \tag{18.2}$$

and by linearity,

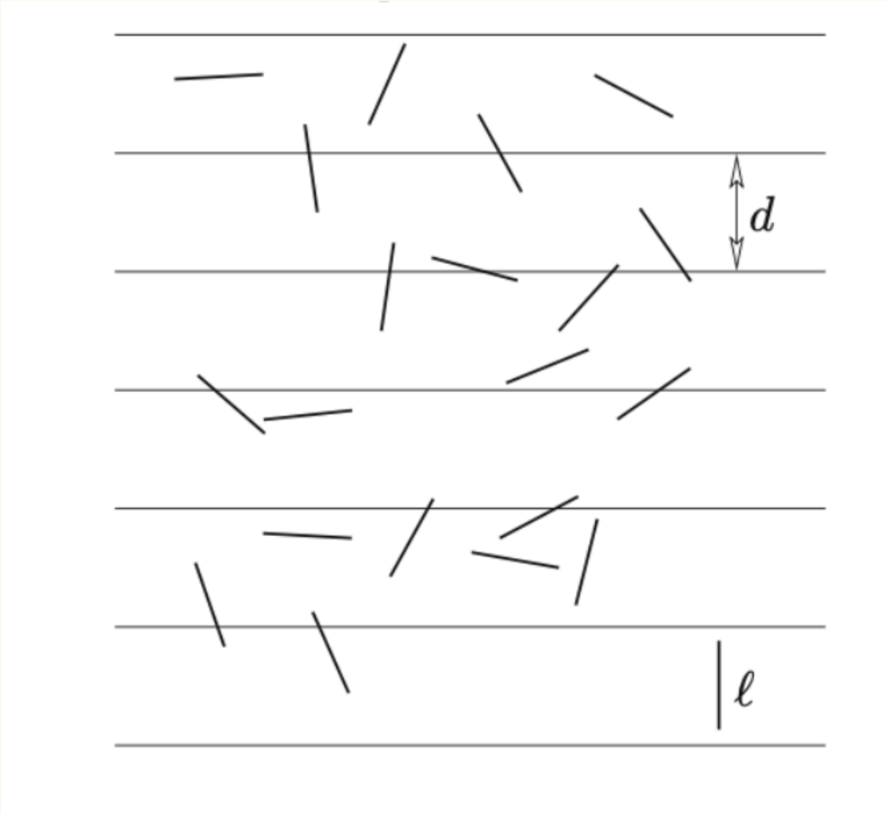
$$\mathbb{E}[X] = \sum_{k \geq 0} \mathbb{E}[I_k] = \sum_{k \geq 0} \Pr(X > k) \tag{18.3}$$

Method 2:

$$\begin{aligned} \mathbb{E}[X] &= \sum_{x \geq 0} x \Pr(X = x) = \sum_{x \geq 0} \sum_{k=0}^{x-1} \Pr(X = x) \\ &= \sum_{k \geq 0} \sum_{x > k} \Pr(X = x) = \sum_{k > 0} \Pr(X > k) \end{aligned} \tag{18.4}$$

5.19 Applications of Linearity

Unsuccessful Search in Open Addressing



Hash Function $h : U \to [m]$, n keys from the universe U are mapped into m slots.

Open Addressing hash collision is resolved by a probing strategy. When searching for a a key $x \in U$, the i-th probed slot is given by $h(x, i)$.

- Linear probing: $h(x, i) = h(x) + i (\operatorname{mod} m)$
- Quadratic probing: $h(x, i) = h(x) + c_1 i + c_2 i^2 (\operatorname{mod} m)$
- Double hashing: $h(x, i) = h_1(x) + i h_2(x) (\operatorname{mod} m)$
- Uniform hashing: $h(x, i) = \pi(i)$ where π is a uniform random permutation of $[m]$.

In a hash table with load factor $\alpha = \frac{n}{m}$, under uniform hashing, let X be the number of probes in an unsuccessful search. Then

$$\mathbb{E}[X] = 1 + \sum_{k \geq 1} \Pr(X > k) \tag{19.1}$$

Let A_i be the event that the i-th probed slot is occupied. By the chain rule,

$$\Pr(X > k) = \Pr\left(\bigcap_{i=1}^{k} A_i\right) = \prod_{i=1}^{k} \Pr\left(A_i \mid \bigcap_{j<i} A_j\right) \tag{19.2}$$

and

$$\Pr\left(A_i \mid \bigcap_{j<i} A_j\right) = \frac{n-i+1}{m-i+1} \leq \alpha \tag{19.3}$$

so

$$\mathbb{E}[X] \leq 1 + \sum_{k\geq 1} \alpha^k = \frac{1}{1-\alpha} \tag{19.4}$$

Indicator Proof of Inclusion-Exclusion

Since

$$I(A^c) = 1 - I(A), \quad I(A \cap B) = I(A)I(B) \tag{19.5}$$

one gets

$$I\left(\bigcup_{i=1}^{n} A_i\right) = 1 - \prod_{i=1}^{n}(1 - I(A_i)) \tag{19.6}$$

Expanding the product yields

$$I\left(\bigcup_{i=1}^{n} A_i\right) = \sum_{\emptyset \neq S \subseteq [n]} (-1)^{|S|-1} I\left(\bigcap_{i\in S} A_i\right) \tag{19.7}$$

and taking **expectations** gives the usual inclusion-exclusion formula.

5.20 Boole-Bonferroni Inequality

For events $A_1, A_2, \ldots, A_n$

$$\begin{aligned} I\left(\bigcup_{i=1}^{n} A_i\right) &= 1 - \prod_{i=1}^{n}(1 - I(A_i)) \\ &= \sum_{\emptyset \neq S \subseteq [n]} (-1)^{|S|-1} I\left(\bigcap_{i\in S} A_i\right) \\ &= \sum_{k=1}^{n} (-1)^{k-1} \sum_{S \in \binom{\{1,\ldots,n\}}{k}} I\left(\bigcap_{i\in S} A_i\right) \end{aligned} \tag{20.1}$$

Observation:

$$X_k := \binom{\sum_{i=1}^n I(A_i)}{k} = \sum_{S \in \binom{\{1,\ldots,n\}}{k}} \prod_{i \in S} I(A_i) = \sum_{S \in \binom{\{1,\ldots,n\}}{k}} I\left(\bigcap_{i \in S} A_i\right) \tag{20.2}$$

X_k在每个$\sum_i I(A_i) = m$下都是$\binom{m}{k}$,利用杨辉三角, 组合数可以被分解为上一行的两个, 偶数最终是上一行头减去尾, 奇数是加上尾.

$$\sum_{k \le 2t} (-1)^{k-1} X_k \le \sum_{k=1}^{n} (-1)^{k-1} X_k \le \sum_{k \le 2t+1} (-1)^{k-1} X_k \tag{20.3}$$

求均值证毕.

5.21 Limitations of Linearity

Note

For infinite sums,

$$\mathbb{E}\left[\sum_{i=1}^{\infty} X_i\right] = \sum_{i=1}^{\infty} \mathbb{E}[X_i] \tag{21.1}$$

requires absolute convergence.

For a random number of random variables,

$$\mathbb{E}\left[\sum_{i=1}^{N} X_i\right] = \mathbb{E}[N]\mathbb{E}[X_1] \tag{21.2}$$

is not true in general without further hypotheses.

5.22 Conditional Expectation

For a discrete random variable X and an event A with $\Pr(A) > 0$, define

$$\mathbb{E}[X \mid A] = \sum_x x \Pr(X = x \mid A) \tag{22.1}$$

assuming absolute convergence.

The conditional pmf is

$$p_{X|A}(x) = \Pr(X = x \mid A) \tag{22.2}$$

and $\mathbb{E}[X \mid A]$ satisfies the usual linearity properties.

5.23 Law of Total Expectation

If $B_1, \ldots, B_n$ is a partition of Ω, then

$$\mathbb{E}[X] = \sum_{i=1}^{n} \mathbb{E}[X \mid B_i] \Pr(B_i) \tag{23.1}$$

Proof

By the law of total probability,

$$\Pr(X = x) = \sum_{i=1}^{n} \Pr(X = x \mid B_i) \Pr(B_i) \tag{23.2}$$

Therefore

$$\begin{aligned}
\mathbb{E}[X] &= \sum_{x} x \Pr(X = x) \\
&= \sum_{x} x \sum_{i=1}^{n} \Pr(X = x \mid B_i) \Pr(B_i) \\
&= \sum_{i=1}^{n} \Pr(B_i) \sum_{x} x \Pr(X = x \mid B_i) \\
&= \sum_{i=1}^{n} \mathbb{E}[X \mid B_i] \Pr(B_i)
\end{aligned} \tag{23.3}$$

5.24 Conditional Expectation Given a Random Variable

For random variables X and Y, the conditional expectation $\mathbb{E}[X \mid Y]$ is the random variable $f(Y)$ where

$$f(y) = \mathbb{E}[X \mid Y = y] \tag{24.1}$$

This extends naturally to $\mathbb{E}[X \mid Y, Z]$ and satisfies the **tower property**

$$\mathbb{E}[\mathbb{E}[X \mid Y]] = \mathbb{E}[X] \tag{24.2}$$

5.25 Conditional Expectation Applications

QuickSort

QSort(A): an array A of n distinct entries

If $n > 1$ then do:

choose a **pivot** $x = A[1]$;

partition A into L with all entries $< x$,

and R with all entries $> x$;

QSort(L) and QSort(R);

Let X_n be the number of comparisons used by QuickSort on a uniform random permutation of n distinct elements, and let

$$t(n) = \mathbb{E}[X_n] \tag{25.1}$$

If B_i is the event that the first element is the i-th smallest, then

$$\begin{aligned} t(n) &= \sum_{i=1}^{n} \mathbb{E}[X_n \mid B_i] \Pr(B_i) \\ &= \frac{1}{n} \sum_{i=1}^{n} \mathbb{E}[n - 1 + X_{i-1} + X_{n-i}] \\ &= n - 1 + \frac{2}{n} \sum_{i=0}^{n-1} t(i) \end{aligned} \tag{25.2}$$

with $t(0) = t(1) = 0$.

Another QuickSort Proof

A is a permutation of $a_1 < a_2 < \ldots < a_n$.

Let X_{ij} indicate whether a_i and a_j are compared during QuickSort.

Observation 1 Each pair are compared at most once. Then

$$X = \sum_{i<j} X_{ij} \tag{25.3}$$

Observation 2 If a_i, a_j are still in the same array, then so are all a_k for $i < k < j$.

Therefore two keys are compared iff iff one of them is chosen as pivot when they are in the same array.

$$\{a_i, a_{i+1}, \ldots, a_j\} \tag{25.4}$$

Hence

$$\mathbb{E}[X_{ij}] = \frac{2}{j-i+1} \tag{25.5}$$

so by linearity

$$\mathbb{E}[X] = \sum_{i<j} \frac{2}{j-i+1} = \sum_{i=1}^{n} \sum_{k=2}^{n-i+1} \frac{2}{k} \leq 2n \sum_{k=1}^{n} \frac{1}{k} = 2n \ln n + O(n) \tag{25.6}$$

Random Family Tree

In a Galton-Watson process with offspring mean

$$\mu = \mathbb{E}[\xi_j^n] \tag{25.7}$$

one has

$$\mathbb{E}[X_n \mid X_{n-1} = k] = \mathbb{E}\left[\sum_{j=1}^{k} \xi_j^{n-1}\right] = k\mu \tag{25.8}$$

so

$$\mathbb{E}[X_n \mid X_{n-1}] = \mu X_{n-1} \tag{25.9}$$

and therefore by the tower property

$$\mathbb{E}[X_n] = \mu\mathbb{E}[X_{n-1}] = \dots = \mu^n \tag{25.10}$$

Consequently,

$$\mathbb{E}\left[\sum_{n=0}^{\infty} X_n\right] = \sum_{n=0}^{\infty} \mu^n \tag{25.11}$$

which equals $\frac{1}{1-\mu}$ if $0 < \mu < 1$, and diverges to $+\infty$ if $\mu \geq 1$.

5.26 Jensen's Inequality

If f is convex(下凸, 如绝对值函数), then

$$\mathbb{E}[f(X)] \geq f(\mathbb{E}[X]) \tag{26.1}$$

If f is concave(凹的, 如对数函数), then

$$\mathbb{E}[f(X)] \leq f(\mathbb{E}[X]) \tag{26.2}$$

5.27 Monotonicity of Expectation

If $X \leq Y$ almost surely, then

$$\mathbb{E}[X] \le \mathbb{E}[Y] \tag{27.1}$$

Proof

Since $X \le Y$ almost surely, we have

$$\Pr(X \le Y) = 1 \tag{27.2}$$

Hence

$$\Pr((X,Y) = (x,y)) = 0, \text{whenever } x > y \tag{27.3}$$

Therefore

$$\begin{aligned} \mathbb{E}[X] &= \sum_x x \Pr(X = x) \\ &= \sum_x x \sum_y \Pr((X,Y) = (x,y)) \\ &= \sum_x x \sum_{y \ge x} \Pr((X,Y) = (x,y)) \end{aligned} \tag{27.4}$$

Since $x \le y$ whenever $y \ge x$, we get

$$\begin{aligned} \mathbb{E}[X] &\le \sum_y y \sum_{x \le y} \Pr((X,Y) = (x,y)) \\ &= \sum_y y \Pr(Y = y) \\ &= \mathbb{E}[Y] \end{aligned} \tag{27.5}$$

In particular, if $X \le c$ almost surely, then

$$\mathbb{E}[X] \le c \tag{27.6}$$

and if $X \ge c$ almost surely, then

$$\mathbb{E}[X] \ge c \tag{27.7}$$

Also, since

$$-|X| \le X \le |X| \tag{27.8}$$

almost surely, monotonicity gives

$$-\mathbb{E}[|X|] \le \mathbb{E}[X] \le \mathbb{E}[|X|] \tag{27.9}$$

and therefore

$$|\mathbb{E}[X]| \le \mathbb{E}[|X|] \tag{27.10}$$

In particular,

$$\mathbb{E}[|X|] \ge |\mathbb{E}[X]| \ge 0 \tag{27.11}$$

5.28 Averaging Principle

If $\Pr(X < c) = 1$, then $\mathbb{E}[X] < c$.

Proof

The assumption $\Pr(X < c) = 1$ means that

$$X < c \tag{28.1}$$

almost surely. In particular,

$$X \le c \tag{28.2}$$

almost surely, so by monotonicity

$$\mathbb{E}[X] \le c \tag{28.3}$$

Contrapositively,

$$\Pr(X \ge \mathbb{E}[X]) > 0 \tag{28.4}$$

Similarly,

$$\Pr(X \le \mathbb{E}[X]) > 0 \tag{28.5}$$

Note

This is the probabilistic method in its simplest form: if there were no outcome with

$$X(\omega) \ge \mathbb{E}[X] \tag{28.6}$$

then $X < \mathbb{E}[X]$ almost surely, contradicting the previous statement.

Therefore there exists an outcome $\omega \in \Omega$ such that

$$X(\omega) \ge \mathbb{E}[X] \tag{28.7}$$

and similarly there exists an outcome such that

$$X(\omega) \le \mathbb{E}[X] \tag{28.8}$$

5.29 Maximum Cut

Randomized Existence Proof

For an undirected graph $G(V, E)$, let each vertex independently join S with probability $\frac{1}{2}$. For each edge $\{u, v\} \in E$, let

$$I_{uv} = I(Y_u \ne Y_v) \tag{29.1}$$

indicate whether the edge crosses the cut. Then

$$|\delta S| = \sum_{\{u,v\} \in E} I_{uv} \tag{29.2}$$

and

$$\mathbb{E}[|\delta S|] = \sum_{\{u,v\} \in E} \Pr(Y_u \neq Y_v) = \frac{|E|}{2} \tag{29.3}$$

Therefore there exists a cut with size at least $\frac{|E|}{2}$.

Moments and Deviations

6.1 Markov Inequality

Let X be a *nonnegative-valued* random variable. Then,

$$\forall a > 0, \Pr(X \geq a) \leq \frac{\mathbb{E}[X]}{a} \tag{1.1}$$

Corollary

$$\forall c > 1, \Pr(X \geq c\mathbb{E}[X]) \leq \frac{1}{c} \tag{1.2}$$

Tight in the worst case

$$\begin{gathered} \forall c > 1, \exists \mu \in \mathbb{R}, \exists X > 0 s.t. \mathbb{E}[X] = \mu \text{ such that} \\ \Pr(X \geq c\mu) = \frac{1}{c} \end{gathered} \tag{1.3}$$

Lower tail variant (sometimes called *reverse Markov's inequality*):

If X has bounded range $X \leq u$, then

$$\Pr(X \leq a) \leq \frac{u - \mathbb{E}[X]}{u - a} \tag{1.4}$$

6.2 Generalized Markov's Inequality

$$\Pr(f(x) \geq a) \leq \frac{\mathbb{E}[f(x)]}{a} \tag{2.1}$$

Observation

$$\begin{gathered}\mu = \mathbb{E}[X], \Pr(|X - \mu| \geq a) \leq ? \\ \Pr(|X - \mu| \geq a) = \Pr((X - \mu)^2 \geq a^2) \leq \frac{\mathbb{E}[(X - \mu)^2]}{a^2}, \text{variance}\end{gathered} \tag{2.2}$$

6.3 Variance and Moments

$\mathbb{E}[X^k]$: kth moment.
$\mathbb{E}[(X - \mathbb{E}[X])^k]$: kth central moment.
$\mathbf{Var}[X] = \mathbb{E}[(X - \mathbb{E}[X])^2]$
standard deviation $\sigma = \sqrt{\mathbf{Var}[X]}$

6.4 Median and Mean

The median of *RV* X is defined to be any value M s.t.

$$\Pr(X \leq m) \geq \frac{1}{2} \text{ and } \Pr(X \geq m) \geq \frac{1}{2} \tag{4.1}$$

$\mu = \mathbb{E}[X]$ is the value that minimizes

$$\mathbb{E}[(X - \mu)^2] \tag{4.2}$$

proof

$$\begin{gathered}f(x) = \mathbb{E}[(X - x)^2] = \mathbb{E}[X^2] - 2x\mathbb{E}[X] + x^2 \text{ is convex} \\ f'(\mu) = 0\end{gathered} \tag{4.3}$$

The median m is the value that minimizes

$$\mathbb{E}[|X - m|] \tag{4.4}$$

proof

$$
\begin{aligned}
& y > m \Rightarrow \Pr(X \geq y) < \frac{1}{2} \\
& \mathbb{E}[|X-y| - |X-m|] \\
= {} & (y-m)\Pr(X \leq m) + \sum_{m<x<y} (m+y-2x)\Pr(X=x) + (m-y)\Pr(X \geq y) \\
\geq {} & \frac{1}{2}(y-m) + \sum_{m<x<y} (m-y)\Pr(X=x) + (m-y)\Pr(X \geq y) \\
\geq {} & \frac{1}{2}(y-m) + \frac{1}{2}(m-y) = 0
\end{aligned}
\tag{4.5}
$$

X is a *RV*, m is median, μ is $\mathbb{E}[X]$, standard deviation is σ then

$$|\mu - m| \leq \sigma \tag{4.6}$$

proof

$$
\begin{aligned}
|\mu - m| &= |\mathbb{E}[X] - m| \\
&< \mathbb{E}[|X-m|], \text{Jensen's Ineq} \\
&\leq E[\![X-\mu]\!] \\
&= \mathbb{E}\left[\sqrt{(X-\mu)^2}\right] \\
&\leq \sqrt{E[(X-\mu)^2]}, \text{Jensen's Ineq} \\
&= \sigma
\end{aligned}
\tag{4.7}
$$

6.5 Variance of Linear Function

$$\mathbf{Var}[X] = \mathbb{E}[(X-\mathbb{E}[X])^2] = \mathbb{E}[X^2] - (\mathbb{E}[X])^2 \tag{5.1}$$

$$\mathbf{Var}[X+Y] = \mathbf{Var}[X] + \mathbf{Var}[Y] + 2(\mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y]), \mathbf{Cov}(X,Y) \tag{5.2}$$

6.6 Covariance

$$\mathbf{Cov}(X,Y) = \mathbb{E}[(X-\mathbb{E}[X])(Y-\mathbb{E}[Y])] = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y] \tag{6.1}$$

Properties:

$$\mathbf{Var}[X] = \mathbf{Cov}(X,X) \tag{6.2}$$

symmetric

$$\mathbf{Cov}(X,Y) = \mathbf{Cov}(Y,X) \tag{6.3}$$

Distributive

$$
\begin{aligned}
\mathbf{Cov}(X+Y,Z) &= \mathbf{Cov}(X,Z) + \mathbf{Cov}(Y,Z) \\
\mathbf{Cov}(aX,Y) &= a\ \mathbf{Cov}(X,Y)
\end{aligned}
\tag{6.4}
$$

If X, Y are **independent** then

$$
\mathbf{Cov}(X,Y) = 0 \Leftrightarrow \mathbb{E}[X]\mathbb{E}[Y] = \mathbb{E}[XY] \tag{6.5}
$$

If *RV* X and Y are independent, then

$$
\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y] \tag{6.6}
$$

proof

$$
\begin{aligned}
\mathbb{E}[XY] &= \sum_{x,y} xy \Pr(X = x \wedge Y = y) = \sum_{x,y} xy \Pr(X = x) \Pr(Y = y) \\
&= \sum_{x} x \Pr(X = x) \sum_{y} y \Pr(Y = y) \\
&= \mathbb{E}[X]\mathbb{E}[Y]
\end{aligned}
\tag{6.7}
$$

Cauchy-Schwarz

$$
\mathbb{E}[XY]^2 \leq \mathbb{E}[X^2]\mathbb{E}[Y^2] \tag{6.8}
$$

proof.

$$
\begin{aligned}
\mathbb{E}[X^2] &= \mathbb{E}[\mathbb{E}[X^2] \mid Y^2] \\
&= \sum_{i,j} x^2 \Pr(X^2 = i \cap Y^2 = j) \\
&:= \sum_{i,j} x^2 p_{ij}
\end{aligned}
\tag{6.9}
$$

$$
\begin{aligned}
\mathbb{E}[Y^2] &= \mathbb{E}[\mathbb{E}[Y^2] \mid X^2] \\
&= \sum_{i,j} y^2 \Pr(X^2 = i \cap Y^2 = j) \\
&:= \sum_{i,j} y^2 p_{ij}
\end{aligned}
\tag{6.10}
$$

$$
\begin{aligned}
\mathbb{E}[X^2]\mathbb{E}[Y^2] &= \sum_{i,j} x^2 p_{ij} \sum_{i,j} y^2 p_{ij} \\
&\overset{\text{cauchy}}{\geq} \left(\sum_{ij} (p_{ij} xy) \right)^2 \\
&= \mathbb{E}[XY]^2
\end{aligned}
\tag{6.11}
$$

Holder

$$\mathbb{E}[XY] \leq E[\![X|^p]\!]^{\frac{1}{p}} E[\![Y|^q]\!]^{\frac{1}{q}}, \text{when } \frac{1}{p} + \frac{1}{q} = 1 \tag{6.12}$$

6.7 Correlation

The **correlation coefficient** of X and Y is

$$\begin{aligned} \rho(X,Y) &= \frac{\mathbf{Cov}(X,Y)}{\sqrt{\mathbf{Var}[X] \cdot \mathbf{Var}[Y]}} \in [-1,1], \\ &\text{by Cauchy-Schwarz}\left(U = \frac{X - \mathbb{E}[X]}{\sigma(X)}, V = \frac{Y - \mathbb{E}[Y]}{\sigma(Y)}\right) \end{aligned} \tag{7.1}$$

Two *RV* are uncorrelated if $\mathbf{Cov}(X,Y) = 0$
Uncorrelated but not independent, e.g. $X \sim U[-1,1], Y = X^2, \mathbb{E}[X] = 0, \mathbb{E}[XY] = \mathbb{E}[X^3] = 0$

6.8 Variance of Sum

$$\begin{aligned} \mathbf{Var}\left[\sum_i X_i\right] &= \sum_i \mathbf{Var}[X_i] + \sum_{i \neq j} \mathbf{Cov}(X_i, X_j) \\ &\overset{\text{if pairwise independent}}{=} \sum_i \mathbf{Var}[X_i] \end{aligned} \tag{8.1}$$

6.9 Chebyshev's Inequality

$$\Pr(|X - \mathbb{E}[X]| \geq a) \leq \frac{\mathbf{Var}[X]}{a^2} \tag{9.1}$$

Corollary

$\Pr(|X - \mathbb{E}[X]| \geq k\sigma) \leq \frac{1}{k^2}$

Tight in the worst case: $\forall k \geq 1, \forall \mu \in R, \forall \sigma > 0, \exists X(\mathbb{E}[X] = \mu, \mathbf{Var}[X] = \sigma^2)$ such that $\Pr(|X - \mu| \geq k\sigma) = \frac{1}{k^2}$

mean trick

- Let $X_1, \ldots, X_n$ be *i.i.d.* random variables with $E[X_i] = \mu$ and $\mathbf{Var}[X_i] = \sigma^2$.
- Empirical mean: $\overline{X} = \frac{1}{n} \sum_{i=1}^n X_i$

$$E[\overline{X}] = \frac{1}{n}\sum_{i=1}^{n} E[X_i] = \mu \quad \text{and} \quad \mathbf{Var}[\overline{X}] = \frac{1}{n^2}\sum_{i=1}^{n}\mathbf{Var}[X_i] = \frac{\sigma^2}{n} \tag{9.2}$$

- Chebyshev's inequality:

$$\Pr\left(|\overline{X} - \mu| \geq \varepsilon\mu\right) \leq \frac{\mathbf{Var}[\overline{X}]}{\varepsilon^2\mu^2} = \frac{\sigma^2}{\varepsilon^2\mu^2 n} \leq \delta \quad \text{if } n \geq \frac{\sigma^2}{\varepsilon^2\mu^2\delta} \tag{9.3}$$

Balls into Bins

Throwing n balls into n bins, the most heavily loaded bin receives $\Theta\left(\frac{\ln n}{\ln\ln n}\right)$ balls *a.a.s* (asymptotically almost surely), with probability $1 - o(1)$.

- upper bound, see above, union bound.
- lower bound.

Choose $k = 0.99\frac{\ln n}{\ln\ln n}$. To proof $\Pr(\text{max-load} < k) = o(1)$ Let

$$I_i = I(\text{bin-}i \text{ receives extactly } \ k \text{ balls}), X = \sum_i I_i \tag{9.4}$$

$$\Pr(\text{max-load} < k) \leq \Pr(X = 0) \leq \Pr(|X - \mathbb{E}[X]| \geq \mathbb{E}[X]) \leq \frac{\mathbf{Var}[X]}{\mathbb{E}[X]^2} \tag{9.5}$$

$$\mathbb{E}[X] = np = n\binom{n}{k}\left(\frac{1}{n}\right)^k\left(1 - \frac{1}{n}\right)^{n-k} \underset{\text{use } \binom{n}{k} \geq \left(\frac{n}{k}\right)^k}{\geq} \frac{n}{ek^k} = \omega(1) \tag{9.6}$$

$$\begin{aligned}\mathbf{Var}[X] &= \sum_i \mathbf{Var}[I_i] + \sum_{i \neq j}\mathbf{Cov}(I_i, I_j) = \sum_{i,j}\left(E_{[I_i I_j]} - E[I_i]E[I_j]\right) \\ &= \sum_{i,j}\left(E[I_i I_j] - p^2\right) = \sum_i E[I_i^2] + \sum_{i \neq j} E[I_i I_j] - n^2p^2 \\ &= \sum_i E[I_i] + \sum_{i \neq j} E[I_i I_j] - n^2p^2 = np + Y - (np)^2\end{aligned} \tag{9.7}$$

$$\frac{\mathbf{Var}[X]}{\mathbb{E}[X]^2} \leq o(1) + \frac{Y - n^2p^2}{n^2p^2} \tag{9.8}$$

$$\begin{aligned}E[I_i I_j] &= \Pr(\text{both receive exactly } k \text{ balls}) \\ &= \binom{n}{k, k, n-2k} n^{-2k}\left(1 - \frac{2}{n}\right)^{n-2k} \\ &\leq (1 + o(1))p^2\left(\text{as } k \ll \sqrt{n}\right)\end{aligned} \tag{9.9}$$

Cliques in Random Graph

- Fix a constant integer $k \geq 3$. Let X be the number of k-cliques (K_k) in $G \sim G(n,p)$.
- For every distinct $S \subseteq [n]$ of size $|S| = k$, let $I_S = I(K_S \subseteq G)$. Then:

$$X = \sum_{S \in \binom{[n]}{k}} I_S \quad \text{and} \quad \mathbb{E}[I_S] = \Pr(K_S \subseteq G) = p^{\binom{k}{2}} \tag{9.10}$$

- **Linearity of expectation:**

$$\mathbb{E}[X] = \binom{n}{k} p^{\binom{k}{2}} = \Theta\left(n^k p^{\binom{k}{2}}\right) \tag{9.11}$$

$$\mathbb{E}[X] = \Theta\left(n^k p^{\binom{k}{2}}\right) = \begin{cases} o(1) & \text{if } p = o\left(n^{-\frac{2}{k-1}}\right) \overset{\text{Markov}}{\Rightarrow} \Pr(X \geq 1) = o(1) \\ \omega(1) & \text{if } p = \omega\left(n^{-\frac{2}{k-1}}\right) \overset{?}{\Rightarrow} \Pr(X \geq 1) = 1 - o(1) \end{cases} \tag{9.12}$$

Same like previous balls into bins,- **Second Moment Method:** To prove $\Pr(X \geq 1) = 1 - o(1)$, we use Chebyshev's inequality on $X = 0$:

$$\Pr(X = 0) \leq \frac{\mathbf{Var}[X]}{\mathbb{E}[X]^2} = \frac{\sum_{S,T} \mathbf{Cov}(I_S, I_T)}{\mathbb{E}[X]^2} \tag{9.13}$$

- **Variance Decomposition:** We split the variance into the sum of individual variances and covariances of distinct pairs:

$$\begin{aligned} \mathbf{Var}[X] &= \sum_S \mathbf{Var}[I_S] + \sum_{S \neq T} \mathbf{Cov}(I_S, I_T) \\ &\leq \mathbb{E}[X] + \sum_{i=2}^{k-1} \sum_{|S \cap T| = i} \mathbb{E}[I_S I_T] \end{aligned} \tag{9.14}$$

- **Bounding the Variance Ratio:** Dividing by $\mathbb{E}[X]^2$, the ratio is bounded by two parts:

$$\frac{\mathbf{Var}[X]}{\mathbb{E}[X]^2} \leq \frac{1}{\mathbb{E}[X]} + \sum_{i=2}^{k-1} \Theta\left(\frac{1}{n^i p^{\binom{i}{2}}}\right) \tag{9.15}$$

  Since $p = \omega\left(n^{-\frac{2}{k-1}}\right)$, we know $\mathbb{E}[X] \to \infty$, making the first term $\frac{1}{\mathbb{E}[X]} = o(1)$. We now bound the sum.

- **Applying the Threshold Rigorously:** Since $p = \omega\left(n^{-\frac{2}{k-1}}\right)$, there exists some function $c_n \to \infty$ such that $p = c_n n^{-\frac{2}{k-1}}$. Let's examine the denominator of any term $i \in [2, k-1]$:

$$n^i p^{\binom{i}{2}} = n^i \left(c_n n^{-\frac{2}{k-1}}\right)^{\frac{i(i-1)}{2}} = c_n^{\binom{i}{2}} n^{i - \frac{i(i-1)}{k-1}} = c_n^{\binom{i}{2}} n^{\frac{i(k-i)}{k-1}} \tag{9.16}$$

  For $2 \leq i \leq k-1$:

  1. The exponent of n is $\frac{i(k-i)}{k-1} > 0$, so $n^{\frac{i(k-i)}{k-1}} \geq 1$ (it strictly grows).

2. Since $i \geq 2$, the exponent of c_n is $\binom{i}{2} \geq 1$. Because $c_n \to \infty$, we have $c_n^{\binom{i}{2}} \to \infty$.

Thus, the entire denominator $c_n^{\binom{i}{2}} n^{\frac{i(k-i)}{k-1}} \to \infty$, meaning every term in the sum is $\frac{1}{\omega(1)} = o(1)$.

- **Conclusion:** We are summing a finite number $(k-2)$ of $o(1)$ terms, plus the $\frac{1}{\mathbb{E}}[X]$ term which is also $o(1)$.

$$\frac{\mathbf{Var}[X]}{\mathbb{E}[X]^2} = o(1) + \sum_{i=2}^{k-1} o(1) = o(1) \quad \Rightarrow \quad \Pr(X = 0) = o(1) \tag{9.17}$$

A "Threshold Behavior" in Random Graph

Use same method, we can conclude:

For $H(V, E)$ with $k = |V|, m = |E|$ s.t. every subgraph of H has density $\leq \frac{m}{k}$.

$$\Pr(G \text{ contains a subgraph } H) = \begin{cases} o(1) & \text{if } p = o\left(n^{-\frac{k}{m}}\right) \\ 1 - o(1) & \text{if } p = \omega\left(n^{-\frac{k}{m}}\right) \end{cases} \tag{9.18}$$

6.10 Weierstrass Approximation Theorem

Let $f : [0, 1] \to [0, 1]$ be a continuous function. For any $\varepsilon > 0$, there exists a polynominal p such that

$$\sup_{x \in [0,1]} |p(x) - f(x)| \leq \varepsilon \tag{10.1}$$

Proof. Let n be sufficiently large(to be fixed later).

For $x \in [0, 1]$, let $X \sim \frac{1}{n}$ Bin(n, x). Define polynominal p on $x \in [0, 1]$ to be

$$p(x) = E[f(X)] = \sum_{k=0}^{n} f\left(\frac{k}{n}\right) \binom{n}{k} x^k (1-x)^{n-k} \tag{10.2}$$

$$|p(x) - f(x)| = |E[f(X) - f(x)]| \leq E[\![f(X) - f(x)]\!] \tag{10.3}$$

f is continuous on $[0, 1] \Rightarrow \exists \delta > 0 s.t.\ |f(x) - f(y)| \leq \frac{\varepsilon}{2}$ for all $|x - y| \leq \delta$, therefore

$$\begin{aligned} & E[\![f(X) - f(x)]\!] \\ = & E[|f(X) - f(x)| \mid |X - x| \leq \delta] \cdot \Pr(|X - x| \leq \delta) + \\ & E[|f(X) - f(x)| \mid |X - x| > \delta] + \cdot \Pr(|X - x| > \delta) \\ \leq & E\left[\frac{\varepsilon}{2}\right] + |1 - 0| \Pr(|X - x| > \delta) \\ \leq & \frac{\varepsilon}{2} + \frac{x(1-x)}{n\delta^2} \leq \frac{\varepsilon}{2} + \frac{1}{4n\varepsilon^2} \leq \varepsilon, \text{if we choose } n \geq \frac{1}{2\varepsilon\delta^2} \end{aligned} \tag{10.4}$$

6.11 The Moment Problem

Question: Do moments $m_k = E[X^k] < \infty$ for all $k \geq 1$, uniquely identify the distribution of X? In other words, if $E[X^k] = E[Y^k]$ for all $k \geq 1$, are X and Y always identically distributed?

- **If X takes values from a finite set: YES.** We can solve this by setting up a linear system (using the Vandermonde matrix):

$$\begin{pmatrix} x_1 & x_2 & \dots & x_n \\ x_1^2 & x_2^2 & \dots & x_n^2 \\ \vdots & \vdots & \ddots & \vdots \\ x_1^n & x_2^n & \dots & x_n^n \end{pmatrix} \begin{pmatrix} p_1 \\ p_2 \\ \vdots \\ p_n \end{pmatrix} = \begin{pmatrix} m_1 \\ m_2 \\ \vdots \\ m_n \end{pmatrix} \tag{11.1}$$

  Since all x_i are distinct, the matrix is invertible, yielding a unique probability vector p.

- **Moment Generating Function (MGF):** To address the general case, we define the MGF of a random variable X as:

$$M_X(t) = E[e^{tX}] \tag{11.2}$$

  By expanding e^{tX} using the Maclaurin series, we can connect the MGF to the moments:

$$M_X(t) = E[e^{tX}] = \sum_{k \geq 0} \frac{t^k E[X^k]}{k!} \tag{11.3}$$

  - **Convergence:** The MGF $M_X(t)$ is convergent if the moment sequence $E[X^k]$ does not grow too fast.
  - **Moment Extraction:** The k-th moment is given by $E[X^k] = M_X^{(k)}(0)$, where $M_X^{(k)}$ is the k-th derivative of M_X evaluated at $t = 0$.

- **Uniqueness Theorem:** If X and Y have the same moment generating function, specifically if $M_X(t) = M_Y(t)$ for all $t \in (-\varepsilon, \varepsilon)$ for some $\varepsilon > 0$, then X and Y are identically distributed.

6.12 Poisson Tails

$X \sim \text{Pois}(\lambda)$. Tail inequalites for $X \sim \text{Pois}(\lambda)$

$$\begin{aligned} k > \lambda \Rightarrow \Pr(X \geq k) \leq e^{-\lambda} \left(\frac{e\lambda}{k}\right)^k \\ k < \lambda \Rightarrow \Pr(X \leq k) \leq e^{-\lambda} \left(\frac{e\lambda}{k}\right)^k \end{aligned} \tag{12.1}$$

- But use Chebyshev

$$\Pr[|X - \lambda| \geq |k - \lambda|] \leq \frac{\mathbf{Var}[X]}{(k-\lambda)^2} = \frac{\lambda}{(k-\lambda)^2} \tag{12.2}$$

- use MGF

$$M_X(t) = E[e^{tX}] = \sum_{k=o}^{\infty} e^{tk} \frac{e^{\lambda}\lambda^k}{k!} = e^{\lambda(e^t-1)} \sum e^{-\lambda e^t} \frac{(\lambda e^t)^k}{k!} = e^{\lambda(e^t-1)} \quad (12.3)$$

6.13 Cramer-Chernoff Method (Chernoff Bound)

$$M_X(t) = e^{\lambda(e^t-1)} \quad (13.1)$$

$$\begin{aligned} \Pr(X \geq k) = \Pr(e^{tX} \geq e^{tk})(\forall t > 0) > \overset{\text{Markov}}{\leq} \frac{E[e^{tX}]}{e^{tk}} \\ = e^{\lambda(e^t-1)-tk} \overset{f'(t)=0}{\leq} e^{-\lambda}\left(\frac{e\lambda}{k}\right)^k \end{aligned} \quad (13.2)$$

Same, $\Pr(X < k)$, let $t < 0$.

Continuous Random Variable

7.1 Continuous Random Variable and Density

Def continuous random variable

A random variable X is **continuous** if there exists an integrable function

$$f_X : \mathbb{R} \to [0, \infty) \quad (1.1)$$

such that for every $x \in \mathbb{R}$,

$$F_X(x) = \Pr(X \leq x) = \int_{-\infty}^{x} f_X(t)\, \mathrm{d}t \quad (1.2)$$

The function f_X is called a **probability density function** (pdf) of X.

Note

Once X has a density, the CDF F_X is automatically continuous. In particular,

$$\Pr(X = x) = F_X(x) - \lim_{t \to x^-} F_X(t) = 0 \quad (1.3)$$

for every $x \in \mathbb{R}$.

Theorem

A function $f : \mathbb{R} \to \mathbb{R}$ is a density of some continuous random variable iff

$$f(x) \geq 0 \text{ for all } x, \quad \int_{-\infty}^{\infty} f(x)\,\mathrm{d}x = 1 \tag{1.4}$$

In that case, if X has density $f_X = f$, then for every $a < b$,

$$\Pr(a < X \leq b) = \int_a^b f_X(x)\,\mathrm{d}x \tag{1.5}$$

Proof

If X is continuous with density f_X, then clearly $f_X(x) \geq 0$ and

$$\int_{-\infty}^{\infty} f_X(x)\,\mathrm{d}x = \lim_{b \to \infty} F_{X(b)} - \lim_{a \to -\infty} F_{X(a)} = 1 \tag{1.6}$$

Conversely, if $f \geq 0$ and $\int_{-\infty}^{\infty} f(x)\,\mathrm{d}x = 1$, define

$$F(x) = \int_{-\infty}^{x} f(t)\,\mathrm{d}t \tag{1.7}$$

Then F is nondecreasing, right-continuous, and satisfies

$$\lim_{x \to -\infty} F(x) = 0, \quad \lim_{x \to \infty} F(x) = 1 \tag{1.8}$$

so F is a valid CDF.

For the interval formula,

$$\Pr(a < X \leq b) = F_{X(b)} - F_{X(a)} = \int_a^b f_X(x)\,\mathrm{d}x \tag{1.9}$$

Note

The density is not uniquely determined pointwise. If two functions agree except on a set of Lebesgue measure 0, then they define the same interval probabilities, hence the same distribution.

If F_X happens to be differentiable, one usually chooses

$$f_X(x) = F_X'(x) \tag{1.10}$$

Example

If

$$f(x) = cx^2 I_{0 \le x \le 1} \tag{1.11}$$

is to be a density, then we must have

$$1 = \int_0^1 cx^2 \, \mathrm{d}x = \frac{c}{3} \tag{1.12}$$

hence $c = 3$.

For the corresponding random variable X,

$$F_X(x) = 0 \quad \text{for } x < 0 \tag{1.13}$$

$$F_X(x) = \int_0^x 3t^2 \, \mathrm{d}t = x^3 \quad \text{for } 0 \le x \le 1 \tag{1.14}$$

and

$$F_X(x) = 1 \quad \text{for } x > 1 \tag{1.15}$$

7.2 Density and Integration

Probability Density Function

For a continuous random variable, the density value $f_X(x)$ is **not** itself a probability. What is meaningful is the probability of a short interval:

$$\Pr(x < X \le x + h) = \int_x^{x+h} f_X(t) \, \mathrm{d}t \approx f_X(x)h \tag{2.1}$$

when h is small.

So the density tells us how probability mass is distributed locally, not the probability of a single point.

Riemann vs. Lebesgue

The Dirichlet function

$$f(x) = I_{x \in \mathbb{Q}} \tag{2.2}$$

on $[0, 1]$ is the standard warning sign for Riemann integration. Every interval contains both rationals and irrationals, so upper and lower sums never settle down to one value.

From the probabilistic point of view, however, what matters is not the height of the graph at each point, but the size of the set on which a given value is taken. This is exactly what Lebesgue integration is designed to measure.

Def **Borel sigma-field**

The **Borel sigma-field** on $\mathbb{R}$, denoted $\mathcal{B}(\mathbb{R})$, is the smallest sigma-field containing all open intervals.

Elements of $\mathcal{B}(\mathbb{R})$ are called **Borel sets**.

Note

In continuous probability, we usually work on a Borel probability space

$$(\mathbb{R}, \mathcal{B}(\mathbb{R}), \Pr) \tag{2.3}$$

or on a measurable space that carries the same kind of interval information. This is the natural setting for CDFs, densities, and change-of-variable formulas.

Lebesgue Integral

Let (Ω, Σ, μ) be a measure space and let $f : \Omega \to [0, \infty]$ be measurable. One convenient way to define the Lebesgue integral is the layer-cake formula

$$\int_\Omega f \, \mathrm{d}\mu = \int_0^\infty \mu(\{\omega \in \Omega : f(\omega) > t\}) \, \mathrm{d}t \tag{2.4}$$

For a general real-valued measurable function, write

$$f = f^+ - f^- \tag{2.5}$$

where

$$f^+ = \max(f, 0), \quad f^- = \max(-f, 0) \tag{2.6}$$

and define the integral whenever at least one of

$$\int f^+ \, \mathrm{d}\mu, \quad \int f^- \, \mathrm{d}\mu \tag{2.7}$$

is finite.

Example

Some standard pathological examples:

1. A Vitali set is non-measurable, so it cannot be assigned a sensible translation-invariant probability.
2. The Dirichlet function $I_{\mathbb{Q}}$ is Lebesgue integrable on $[0, 1]$ with value 0, because $\mathbb{Q} \cap [0, 1]$ is countable.
3. The Cantor set is uncountable but has Lebesgue measure 0.

These examples explain why measure-theoretic language is not optional decoration here; it is what keeps the theory correct.

7.3 Jointly Continuous Random Variables

Def joint CDF and joint density

For random variables X and Y, their joint distribution function is

$$F_{X,Y}(x, y) = \Pr(X \leq x \wedge Y \leq y) \tag{3.1}$$

They are **jointly continuous** if there exists a function

$$f_{X,Y} : \mathbb{R}^2 \to [0, \infty) \tag{3.2}$$

such that

$$F_{X,Y}(x, y) = \int_{-\infty}^{y} \int_{-\infty}^{x} f_{X,Y}(u, v) \, \mathrm{d}u \, \mathrm{d}v \tag{3.3}$$

for all $x, y \in \mathbb{R}$.

Note

If $F_{X,Y}$ is sufficiently differentiable, then one recovers the joint density by

$$f_{X,Y}(x, y) = \frac{\partial^2 F_{X,Y}(x, y)}{\partial x \, \mathrm{d}y} \tag{3.4}$$

7.4 Marginal Distribution

If (X, Y) has joint density $f_{X,Y}$, then the marginals are obtained by integrating out the other coordinate:

$$f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x, y) \, \mathrm{d}y \tag{4.1}$$

$$f_Y(y) = \int_{-\infty}^{\infty} f_{X,Y}(x, y) \,\mathrm{d}x \tag{4.2}$$

Similarly,

$$F_X(x) = F_{X,Y}(x, \infty), \quad F_Y(y) = F_{X,Y}(\infty, y) \tag{4.3}$$

Proof

For example,

$$F_X(x) = \Pr(X \le x) = \Pr(X \le x \wedge Y \in \mathbb{R}) \tag{4.4}$$

so

$$F_X(x) = \int_{-\infty}^{\infty} \int_{-\infty}^{x} f_{X,Y}(u, y) \,\mathrm{d}u \,\mathrm{d}y \tag{4.5}$$

Differentiating with respect to x gives

$$f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x, y) \,\mathrm{d}y \tag{4.6}$$

7.5 Independence

Two random variables X and Y are independent iff for all $x, y \in \mathbb{R}$,

$$F_{X,Y}(x, y) = F_X(x) F_Y(y) \tag{5.1}$$

If X and Y are jointly continuous, this is equivalent to

$$f_{X,Y}(x, y) = f_X(x) f_Y(y) \tag{5.2}$$

for all x, y.

Proof

If

$$f_{X,Y}(x, y) = f_X(x) f_Y(y) \tag{5.3}$$

then

$$F_{X,Y}(x, y) = \int_{-\infty}^{y} \int_{-\infty}^{x} f_{X(u)} f_{Y(v)} \,\mathrm{d}u \,\mathrm{d}v = F_X(x) F_Y(y) \tag{5.4}$$

Conversely, if

$$F_{X,Y}(x, y) = F_X(x) F_Y(y) \tag{5.5}$$

and the functions are differentiable enough, differentiating in x and y yields

$$f_{X,Y}(x, y) = f_X(x) f_Y(y) \tag{5.6}$$

Note

If X and Y are independent and g, h are Borel-measurable, then $g(X)$ and $h(Y)$ are also independent.

7.6 Conditional Distribution

Def conditioning on an event

Let X be continuous and let A be an event with $\Pr(A) > 0$. The conditional CDF of X given A is

$$F_{X|A}(x) = \Pr(X \leq x \mid A) \tag{6.1}$$

If this conditional law has a density, we denote it by $f_{X|A}$, so that

$$F_{X|A}(x) = \int_{-\infty}^{x} f_{X|A}(u) \, \mathrm{d}u \tag{6.2}$$

Law of Total Probability for Densities

If $B_1, \ldots, B_n$ is a partition of Ω and each $\Pr(B_i) > 0$, then

$$f_X(x) = \sum_{i=1}^{n} \Pr(B_i) f_{X|B_i}(x) \tag{6.3}$$

Proof. For each x,

$$\Pr(X \leq x) = \sum_{i=1}^{n} \Pr(B_i) \Pr(X \leq x \mid B_i) \tag{6.4}$$

Differentiating with respect to x gives the density identity.

Def conditional density given a random variable

Suppose (X, Y) is jointly continuous. For every y with $f_Y(y) > 0$, define

$$F_{X|Y}(x \mid y) = \Pr(X \leq x \mid Y = y) \tag{6.5}$$

by

$$F_{X|Y}(x \mid y) = \int_{-\infty}^{x} \frac{f_{X,Y}(u,y)}{f_Y(y)}\,\mathrm{d}u \tag{6.6}$$

Hence the conditional density is

$$f_{X|Y}(x \mid y) = \frac{f_{X,Y}(x,y)}{f_Y(y)} \tag{6.7}$$

Proof

Fix y and a very short interval $(y, y + \mathrm{d}y]$. Then heuristically

$$\Pr(X \le x \mid y < Y \le y + \mathrm{d}y) = \frac{\Pr(X \le x \land y < Y \le y + \mathrm{d}y)}{\Pr(y < Y \le y + \mathrm{d}y)} \tag{6.8}$$

Using the joint density,

$$\Pr(X \le x \land y < Y \le y + \mathrm{d}y) \approx \int_{-\infty}^{x} f_{X,Y}(u,y)\,\mathrm{d}u \cdot \mathrm{d}y \tag{6.9}$$

while

$$\Pr(y < Y \le y + \mathrm{d}y) \approx f_Y(y)\,\mathrm{d}y \tag{6.10}$$

Cancelling the common factor $\mathrm{d}y$ gives

$$F_{X|Y}(x \mid y) = \int_{-\infty}^{x} \frac{f_{X,Y}(u,y)}{f_Y(y)}\,\mathrm{d}u \tag{6.11}$$

Law of Total Probability Given a Continuous Variable

Let $B \subseteq \mathbb{R}$ be a Borel set. Then

$$\Pr(X \in B) = \int_{-\infty}^{\infty} \Pr(X \in B \mid Y = y) f_Y(y)\,\mathrm{d}y \tag{6.12}$$

In density form, this is just

$$f_X(x) = \int_{-\infty}^{\infty} f_{X|Y}(x \mid y) f_Y(y)\,\mathrm{d}y \tag{6.13}$$

7.7 Expectation and Moments

Def **expectation**

Let X be a continuous random variable with density f_X. Its expectation is

$$\mathbb{E}[X] = \int_{-\infty}^{\infty} x f_X(x)\,\mathrm{d}x = \int_{-\infty}^{\infty} x\,\mathrm{d}F_X(x) \tag{7.1}$$

whenever the integral exists absolutely.

More generally, the k-th moment is

$$\mathbb{E}[X^k] = \int_{-\infty}^{\infty} x^k f_X(x)\,\mathrm{d}x \tag{7.2}$$

Note

The integral notation

$$\int_{-\infty}^{\infty} x\,\mathrm{d}F_X(x) \tag{7.3}$$

is the Stieltjes form. For continuous random variables this agrees with the usual density formula.

7.8 Tail Integral Formula

Theorem

If $X \geq 0$ almost surely, then

$$\mathbb{E}[X] = \int_0^{\infty} \Pr(X > x)\,\mathrm{d}x = \int_0^{\infty} (1 - F_X(x))\,\mathrm{d}x \tag{8.1}$$

Proof

Since $X \geq 0$,

$$\mathbb{E}[X] = \int_0^{\infty} x f_X(x)\,\mathrm{d}x \tag{8.2}$$

and

$$1 - F_X(x) = \int_x^{\infty} f_{X(u)}\,\mathrm{d}u \tag{8.3}$$

Therefore

$$\int_0^{\infty} (1 - F_X(x))\,\mathrm{d}x = \int_0^{\infty} \int_x^{\infty} f_{X(u)}\,\mathrm{d}u\,\mathrm{d}x \tag{8.4}$$

The integration region is

$$\{(x,u) : 0 \le x \le u < \infty\} \tag{8.5}$$

so switching the order gives

$$\int_0^\infty \int_0^u f_{X(u)}\,\mathrm{d}x\,\mathrm{d}u = \int_0^\infty u f_{X(u)}\,\mathrm{d}u = \mathbb{E}[X] \tag{8.6}$$

7.9 LOTUS

Law of the Unconscious Statistician

If X is continuous and $g(X)$ is integrable, then

$$\mathbb{E}[g(X)] = \int_{-\infty}^{\infty} g(x) f_X(x)\,\mathrm{d}x \tag{9.1}$$

Proof

First assume $g \ge 0$. By the tail-integral formula,

$$\mathbb{E}[g(X)] = \int_0^\infty \Pr(g(X) > y)\,\mathrm{d}y \tag{9.2}$$

Let

$$B_y = \{x \in \mathbb{R} : g(x) > y\} \tag{9.3}$$

Then

$$\Pr(g(X) > y) = \int_{B_y} f_X(x)\,\mathrm{d}x \tag{9.4}$$

so

$$\mathbb{E}[g(X)] = \int_0^\infty \int_{B_y} f_X(x)\,\mathrm{d}x\,\mathrm{d}y = \int_{-\infty}^{\infty} f_X(x) \int_0^{g(x)} \mathrm{d}y\,\mathrm{d}x = \int_{-\infty}^{\infty} g(x) f_X(x)\,\mathrm{d}x \tag{9.5}$$

For a general g, write

$$g = g^+ - g^- \tag{9.6}$$

and apply the nonnegative case to g^+ and g^- separately.

7.10 Linearity, Monotonicity, and Total Expectation

For scalars a, b and integrable random variables X, Y,

$$\mathbb{E}[aX + b] = a\mathbb{E}[X] + b \tag{10.1}$$

$$\mathbb{E}[X + Y] = \mathbb{E}[X] + \mathbb{E}[Y] \tag{10.2}$$

Proof

The first identity is immediate:

$$\mathbb{E}[aX + b] = \int_{-\infty}^{\infty} (ax + b) f_X(x)\,\mathrm{d}x = a \int_{-\infty}^{\infty} x f_X(x)\,\mathrm{d}x + b \int_{-\infty}^{\infty} f_X(x)\,\mathrm{d}x = a\mathbb{E}[X] + b \tag{10.3}$$

For the second, use the joint density:

$$\mathbb{E}[X + Y] = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} (x + y) f_{X,Y}(x, y)\,\mathrm{d}x\,\mathrm{d}y \tag{10.4}$$

$$= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x f_{X,Y}(x, y)\,\mathrm{d}x\,\mathrm{d}y + \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} y f_{X,Y}(x, y)\,\mathrm{d}x\,\mathrm{d}y = \mathbb{E}[X] + \mathbb{E}[Y] \tag{10.5}$$

If $X \leq Y$ almost surely, then

$$\mathbb{E}[X] \leq \mathbb{E}[Y] \tag{10.6}$$

Proof

Let

$$Z = Y - X \tag{10.7}$$

Then $Z \geq 0$ almost surely, so by nonnegativity of the integral,

$$\mathbb{E}[Z] \geq 0 \tag{10.8}$$

Hence

$$\mathbb{E}[Y] - \mathbb{E}[X] = \mathbb{E}[Z] \geq 0 \tag{10.9}$$

Total Expectation

If $B_1, \ldots, B_n$ is a partition of Ω with $\Pr(B_i) > 0$, then

$$\mathbb{E}[X] = \sum_{i=1}^{n} \mathbb{E}[X \mid B_i] \Pr(B_i) \tag{10.10}$$

If (X, Y) is jointly continuous, then

$$\mathbb{E}[X] = \int_{-\infty}^{\infty} \mathbb{E}[X \mid Y = y] f_Y(y) \, \mathrm{d}y \tag{10.11}$$

and equivalently

$$\mathbb{E}[\mathbb{E}[X \mid Y]] = \mathbb{E}[X] \tag{10.12}$$

7.11 Expectation of Products

If X and Y are independent and integrable, then

$$\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y] \tag{11.1}$$

Proof

Independence gives

$$f_{X,Y}(x, y) = f_X(x) f_Y(y) \tag{11.2}$$

therefore

$$\mathbb{E}[XY] = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} xy f_{X,Y}(x, y) \, \mathrm{d}x \, \mathrm{d}y = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} xy f_X(x) f_Y(y) \, \mathrm{d}x \, \mathrm{d}y \tag{11.3}$$

$$= \int_{-\infty}^{\infty} x f_X(x) \, \mathrm{d}x \cdot \int_{-\infty}^{\infty} y f_Y(y) \, \mathrm{d}y = \mathbb{E}[X]\mathbb{E}[Y] \tag{11.4}$$

Note

Consequently,

$$\mathbf{Cov}(X, Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y] = 0 \tag{11.5}$$

whenever X and Y are independent.

More generally, if $X_1, \ldots, X_n$ are pairwise independent, then

$$\mathbf{Var}\left(\sum_{i=1}^{n} X_i\right) = \sum_{i=1}^{n} \mathbf{Var}(X_i) \tag{11.6}$$

7.12 Uniform Distribution

Def uniform distribution

The random variable X is **uniform on** $[a, b]$ with $a < b$ if it has density

$$f_X(x) = \frac{1}{b-a} \quad \text{for } a \le x \le b \tag{12.1}$$

$$f_X(x) = 0 \quad \text{for } x < a \text{ or } x > b \tag{12.2}$$

Its CDF is

$$F_X(x) = 0 \quad \text{if } x < a \tag{12.3}$$

$$F_X(x) = \frac{x-a}{b-a} \quad \text{if } a \le x \le b \tag{12.4}$$

$$F_X(x) = 1 \quad \text{if } x > b \tag{12.5}$$

Note

There is no uniform distribution on an infinite interval such as $[a, \infty)$ or on the whole line $\mathbb{R}$, because a constant density over an infinite region cannot integrate to 1.

Moments of the Uniform Law

If X is uniform on $[a, b]$, then

$$\mathbb{E}[X] = \int_a^b \frac{x}{b-a}\,\mathrm{d}x = \frac{a+b}{2} \tag{12.6}$$

and

$$\mathbb{E}[X^2] = \int_a^b \frac{x^2}{b-a}\,\mathrm{d}x = \frac{a^2+ab+b^2}{3} \tag{12.7}$$

Hence

$$\mathbf{Var}(X) = \mathbb{E}[X^2] - \mathbb{E}[X]^2 = \frac{(b-a)^2}{12} \tag{12.8}$$

Note

The mean is the midpoint of the interval, just as in the discrete uniform case, but the variance is different because the mass is spread continuously rather than sitting on finitely many atoms.

7.13 Rejection Sampling as Conditioning

Uniform Law on a Subinterval

Let X be uniform on $[a, b]$, and let $[c, d] \subseteq [a, b]$. Then

$$\Pr(X \in [c, d]) = \frac{d-c}{b-a} \tag{13.1}$$

Conditioned on the event $X \in [c, d]$, the random variable X is uniform on $[c, d]$.

Proof

For $x \in \mathbb{R}$,

$$\Pr(X \leq x \mid X \in [c, d]) = \frac{\Pr(X \in [c, d] \cap X \leq x)}{\Pr(X \in [c, d])} \tag{13.2}$$

This is

$$0 \quad \text{if } x < c \tag{13.3}$$

$$\frac{x-c}{d-c} \quad \text{if } c \leq x \leq d \tag{13.4}$$

and

$$1 \quad \text{if } x > d \tag{13.5}$$

which is exactly the CDF of the uniform law on $[c, d]$.

7.14 Function of a Random Variable

Monotone Change of Variables

Let $Y = g(X)$ where X has density f_X.

If g is strictly increasing, then

$$F_Y(y) = \Pr(Y \leq y) = \Pr(g(X) \leq y) = \Pr(X \leq g^{-1}(y)) = F_{X(g^{-1}(y))} \tag{14.1}$$

Hence

$$f_Y(y) = f_{X(g^{-1}(y))} \frac{\mathrm{d}g^{-1}(y)}{\mathrm{d}y} \tag{14.2}$$

If g is strictly decreasing, the same computation gives the absolute-value form

$$f_Y(y) = f_{X(g^{-1}(y))} \left| \frac{\mathrm{d}g^{-1}(y)}{\mathrm{d}y} \right| \tag{14.3}$$

Example

If X is continuous and $Y = aX + b$ with $a \neq 0$, then

$$f_Y(y) = \frac{1}{|a|} f_{X(\frac{y-b}{a})} \tag{14.4}$$

In particular, scaling stretches the density by the reciprocal factor $\frac{1}{|a|}$.

7.15 Inverse Transform Sampling

Simulation from a CDF

Let U be uniform on $[0, 1]$ and let F be a continuous CDF. Define

$$X = F^{-1}(U) \tag{15.1}$$

Then X has CDF F.

Proof

For every $x \in \mathbb{R}$,

$$\Pr(X \leq x) = \Pr(F^{-1}(U) \leq x) = \Pr(U \leq F(x)) \tag{15.2}$$

because F^{-1} is increasing.

Since U is uniform on $[0, 1]$,

$$\Pr(U \leq F(x)) = F(x) \tag{15.3}$$

so indeed X has CDF F.

Note

For an integer-valued discrete CDF F, the analogous construction is

$$X = k \quad \text{iff } F(k-1) < U \leq F(k) \tag{15.4}$$

This is the quantile viewpoint: $F^{-1}(u)$ picks the u-th quantile.

7.16 Stochastic Domination and Coupling

Def stochastic domination

Suppose X and Y are real-valued random variables. We say that X **stochastically dominates** Y if

$$F_X(t) \leq F_Y(t) \quad \text{for all } t \in \mathbb{R} \tag{16.1}$$

Theorem

The following are equivalent:

1. X stochastically dominates Y.
2. There exists a coupling $(\sim X, \sim Y)$ with marginals distributed as X and Y respectively such that

$$\sim X \geq \sim Y \tag{16.2}$$

almost surely.

Proof

The implication $2 \Rightarrow 1$ is immediate: if $\sim X \geq \sim Y$ almost surely, then

$$\{\sim X \leq t\} \subseteq \{\sim Y \leq t\} \tag{16.3}$$

for every t, hence

$$F_X(t) = \Pr(\sim X \leq t) \leq \Pr(\sim Y \leq t) = F_Y(t) \tag{16.4}$$

For $1 \Rightarrow 2$, let U be uniform on $[0, 1]$ and define

$$\sim X = F_X^{-1}(U), \quad \sim Y = F_Y^{-1}(U) \tag{16.5}$$

By inverse transform sampling, these have the correct marginals.

Because $F_X(t) \leq F_Y(t)$ for all t, the quantile function of X lies above that of Y, hence

$$\sim X \geq \sim Y \tag{16.6}$$

almost surely.

Note

Coupling is often the cleanest way to compare random variables. Instead of comparing formulas for their CDFs, one puts them on the same probability space and compares them pointwise.

7.17 Exponential Distribution

Def exponential distribution

A random variable X has the **exponential distribution** with parameter $\lambda > 0$, written

$$X \sim \mathrm{Exp}(\lambda) \tag{17.1}$$

if its density is

$$f_X(x) = \lambda \exp(-\lambda x) \quad \text{for } x \geq 0 \tag{17.2}$$

$$f_X(x) = 0 \quad \text{for } x < 0 \tag{17.3}$$

Its CDF is

$$F_X(x) = 0 \quad \text{for } x < 0 \tag{17.4}$$

$$F_X(x) = 1 - \exp(-\lambda x) \quad \text{for } x \geq 0 \tag{17.5}$$

Continuous Limit of the Geometric Law

Think of Bernoulli trials being performed every $\delta > 0$ units of time, with success probability

$$p = \lambda \delta \tag{17.6}$$

at each trial.

Let T_δ be the waiting time for the first success. Then for $x \geq 0$,

$$\Pr(T_\delta > x) = (1 - \lambda\delta)^{\left[\frac{x}{\delta}\right]} \to \exp(-\lambda x) \tag{17.7}$$

as $\delta \to 0$.

So the exponential law is the continuous-time analogue of the geometric law.

Moments of the Exponential Law

If $X \sim \mathrm{Exp}(\lambda)$, then

$$\mathbb{E}[X] = \int_0^\infty x \lambda \exp(-\lambda x)\, \mathrm{d}x = \frac{1}{\lambda} \tag{17.8}$$

One can also use the tail formula:

$$\mathbb{E}[X] = \int_0^\infty \Pr(X > x)\, \mathrm{d}x = \int_0^\infty \exp(-\lambda x)\, \mathrm{d}x = \frac{1}{\lambda} \tag{17.9}$$

Moreover,

$$\mathbb{E}[X^2] = \int_0^\infty x^2 \lambda \exp(-\lambda x)\, \mathrm{d}x = \frac{2}{\lambda^2} \tag{17.10}$$

hence

$$\mathbf{Var}(X) = \mathbb{E}[X^2] - \mathbb{E}[X]^2 = \frac{1}{\lambda^2} \tag{17.11}$$

Theorem

The exponential distribution is memoryless: for all $s, t \geq 0$,

$$\Pr(X > s + t \mid X > t) = \Pr(X > s) \tag{17.12}$$

Proof

Since

$$\Pr(X > u) = \exp(-\lambda u) \tag{17.13}$$

for every $u \geq 0$,

$$\Pr(X > s + t \mid X > t) = \frac{\Pr(X > s + t)}{\Pr(X > t)} = \frac{\exp(-\lambda(s + t))}{\exp(-\lambda t)} = \exp(-\lambda s) = \Pr(X > s) \tag{17.14}$$

Note

In fact, the exponential law is the only continuous distribution on $[0, \infty)$ with the memoryless property.

Minimum of Independent Exponentials

If $X_1, ..., X_n$ are independent and

$$X_i \sim \text{Exp}(\lambda_i) \tag{17.15}$$

then

$$\min(X_1, ..., X_n) \sim \text{Exp}(\lambda_1 + ... + \lambda_n) \tag{17.16}$$

Proof

For $x \geq 0$,

$$\Pr(\min(X_1, ..., X_n) > x) = \Pr(X_1 > x \wedge ... \wedge X_n > x) \tag{17.17}$$

By independence,

$$= \prod_{i=1}^{n} \Pr(X_i > x) = \prod_{i=1}^{n} \exp(-\lambda_i x) = \exp(-(\lambda_1 + ... + \lambda_n)x) \tag{17.18}$$

which is exactly the tail of an exponential distribution with parameter

$$\lambda_1 + \ldots + \lambda_n \tag{17.19}$$

7.18 Poisson Point Process

Poisson Process

A **Poisson process** with rate $\lambda > 0$ is a continuous-time counting process

$$\{N(t) : t \geq 0\} \tag{18.1}$$

defined by the picture of a random clock:

1. $N(t)$ counts how many times the clock has rung by time t, with

$$N(0) = 0 \tag{18.2}$$

2. The interarrival times between consecutive rings, including the waiting time before the first ring, are i.i.d.

$$\operatorname{Exp}(\lambda) \tag{18.3}$$

Note

The exponential memoryless property makes the process restart after every ring. That is why the Poisson process is the natural continuous-time analogue of repeated Bernoulli trials.

Superposition of Clocks

If we have k independent exponential clocks, each with rate λ, then the first ring among all clocks occurs after an

$$\operatorname{Exp}(k\lambda) \tag{18.4}$$

waiting time.

So the process obtained by superposing k independent rate-λ clocks behaves like a single clock of rate $k\lambda$.

Theorem

For every $s, t \geq 0$ and integer $n \geq 0$,

$$\Pr(N(t+s) - N(s) = n) = \exp(-\lambda t)\frac{(\lambda t)^n}{n!} \tag{18.5}$$

In particular,

$$N(t) \sim \mathrm{Pois}(\lambda t) \tag{18.6}$$

Proof

Write

$$p_{n(t)} = \Pr(N(t) = n) \tag{18.7}$$

For a very short time interval of length h, we have

$$\Pr(N(h) = 0) = 1 - \lambda h + o(h) \tag{18.8}$$

$$\Pr(N(h) = 1) = \lambda h + o(h) \tag{18.9}$$

$$\Pr(N(h) \geq 2) = o(h) \tag{18.10}$$

Therefore

$$p_0(t+h) = p_0(t)(1 - \lambda h) + o(h) \tag{18.11}$$

so

$$p_0'(t) = -\lambda p_0(t), \quad p_0(0) = 1 \tag{18.12}$$

and hence

$$p_0(t) = \exp(-\lambda t) \tag{18.13}$$

For $n \geq 1$,

$$p_{n(t+h)} = p_{n(t)}(1 - \lambda h) + p_{n-1}(t)\lambda h + o(h) \tag{18.14}$$

which gives

$$p_{n(t)}' = -\lambda p_{n(t)} + \lambda p_{n-1}(t), \quad p_{n(0)} = 0 \tag{18.15}$$

One checks by induction that the solution is

$$p_{n(t)} = \exp(-\lambda t)\frac{(\lambda t)^n}{n!} \tag{18.16}$$

By the memoryless property, after time s the process restarts afresh, so the same formula holds for the increment

$$N(t+s) - N(s) \tag{18.17}$$

7.19 Normal Distribution

Def **normal distribution**

A continuous random variable X has the **normal** or **Gaussian** distribution with parameters $\mu \in \mathbb{R}$ and $\sigma^2 > 0$, written

$$X \sim \mathrm{N}(\mu, \sigma^2) \tag{19.1}$$

if

$$f_X(x) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right), \quad -\infty < x < \infty \tag{19.2}$$

Standard Normal

The case

$$\mu = 0, \quad \sigma = 1 \tag{19.3}$$

is called the **standard normal distribution**, denoted

$$\mathrm{N}(0, 1) \tag{19.4}$$

with density

$$\varphi(x) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{x^2}{2}\right) \tag{19.5}$$

and CDF

$$\Phi(z) = \int_{-\infty}^{z} \varphi(x)\, \mathrm{d}x \tag{19.6}$$

Note

The CDF Φ has no elementary closed form. One often uses numerical tables or the error function

$$\mathrm{erf}(z) = \frac{2}{\sqrt{\pi}} \int_0^z \exp(-x^2)\, \mathrm{d}x \tag{19.7}$$

and then

$$\Phi(z) = \frac{1}{2} + \frac{1}{2}\, \mathrm{erf}\left(\frac{z}{\sqrt{2}}\right) \tag{19.8}$$

Why This Is a Probability Distribution

The normal density integrates to 1 because of the Gaussian integral

$$\int_{-\infty}^{\infty} \exp\left(-\frac{x^2}{2}\right) \mathrm{d}x = \sqrt{2\pi} \tag{19.9}$$

This is the normalization constant behind

$$\frac{1}{\sqrt{2\pi}} \tag{19.10}$$

in the standard normal density.

Note

The normal law is central because it appears as a continuous limit of the binomial law and, much more generally, through the central limit theorem: sums of many independent random contributions are approximately normal.

Theorem

If

$$X \sim \mathrm{N}(\mu, \sigma^2) \tag{19.11}$$

then

$$\mathbb{E}[X] = \mu, \quad \mathbf{Var}(X) = \sigma^2 \tag{19.12}$$

Proof

Because the density is symmetric around μ, the odd part of

$$x - \mu \tag{19.13}$$

integrates to 0, so

$$\mathbb{E}[X] = \mu \tag{19.14}$$

For the variance, substitute

$$y = \frac{x - \mu}{\sigma} \tag{19.15}$$

Then

$$\mathbf{Var}(X) = \mathbb{E}[(X-\mu)^2] = \int_{-\infty}^{\infty} (x-\mu)^2 \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right) \mathrm{d}x \tag{19.16}$$

$$= \sigma^2 \int_{-\infty}^{\infty} y^2 \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{y^2}{2}\right) \mathrm{d}y \tag{19.17}$$

Integrating by parts gives

$$\int_{-\infty}^{\infty} y^2 \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{y^2}{2}\right) \mathrm{d}y = 1 \tag{19.18}$$

so

$$\mathbf{Var}(X) = \sigma^2 \tag{19.19}$$

7.20 Linear Transformation of a Normal Variable

Theorem

If

$$X \sim \mathrm{N}(\mu, \sigma^2) \tag{20.1}$$

and $a \neq 0$, then for every $b \in \mathbb{R}$,

$$aX + b \sim \mathrm{N}(a\mu + b, a^2\sigma^2) \tag{20.2}$$

Proof

Let

$$Y = aX + b \tag{20.3}$$

Since $y = ax + b$ is monotone, the change-of-variable formula gives

$$f_Y(y) = \frac{1}{|a|} f_{X\left(\frac{y-b}{a}\right)} \tag{20.4}$$

Hence

$$f_Y(y) = \frac{1}{\sqrt{2\pi}|a|\sigma} \exp\left(-\frac{\left(\left(\frac{y-b}{a}\right) - \mu\right)^2}{2\sigma^2}\right) \tag{20.5}$$

$$= \frac{1}{\sqrt{2\pi}\sqrt{a^2\sigma^2}} \exp\left(-\frac{(y - (a\mu + b))^2}{2a^2\sigma^2}\right) \tag{20.6}$$

which is exactly the density of

$$\mathrm{N}(a\mu + b, a^2\sigma^2) \tag{20.7}$$

Note

In particular:

$$X \sim \mathrm{N}(\mu, \sigma^2) \Rightarrow \frac{X - \mu}{\sigma} \sim \mathrm{N}(0, 1) \tag{20.8}$$

and conversely, if $Z \sim \mathrm{N}(0, 1)$, then

$$\sigma Z + \mu \sim \mathrm{N}(\mu, \sigma^2) \tag{20.9}$$

7.21 Sum of Independent Normal Variables

Convolution

For continuous independent random variables X and Y,

$$f_{X+Y}(z) = \int_{-\infty}^{\infty} f_X(x) f_{Y(z-x)} \,\mathrm{d}x \tag{21.1}$$

This is the continuous analogue of convolution from the discrete case.

Theorem

If

$$X \sim \mathrm{N}(\mu, \sigma^2), \quad Y \sim \mathrm{N}(\nu, \tau^2) \tag{21.2}$$

are independent, then

$$X + Y \sim \mathrm{N}(\mu + \nu, \sigma^2 + \tau^2) \tag{21.3}$$

Proof

One way is by convolution: substituting the two Gaussian densities and completing the square shows that

$$f_{X+Y}(z) \tag{21.4}$$

is again Gaussian with mean $\mu + \nu$ and variance $\sigma^2 + \tau^2$.

A quicker route is via moment generating functions, which we derive next.

7.22 Moment Generating Function

Def moment generating function

The **moment generating function** of a random variable X is

$$M_X(t) = \mathbb{E}[\exp(tX)] \tag{22.1}$$

whenever this expectation is finite.

Note

Expanding $\exp(tX)$ into its Maclaurin series suggests

$$M_X(t) = \sum_{k \geq 0} \mathbb{E}[X^k] \frac{t^k}{k!} \tag{22.2}$$

so the moments are recovered by differentiation:

$$\mathbb{E}[X^k] = M_X^k(0) \tag{22.3}$$

If two random variables have the same MGF on an open interval around 0, then they have the same distribution.

MGF of the Standard Normal

If $Z \sim \mathrm{N}(0, 1)$, then

$$M_{Z(t)} = \exp\left(\frac{t^2}{2}\right) \tag{22.4}$$

Proof

Compute directly:

$$M_{Z(t)} = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \exp(tx) \exp\left(-\frac{x^2}{2}\right) \mathrm{d}x \tag{22.5}$$

$$= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \exp\left(-\frac{x^2 - 2tx}{2}\right) \mathrm{d}x \tag{22.6}$$

Complete the square:

$$x^2 - 2tx = (x - t)^2 - t^2 \tag{22.7}$$

Therefore

$$M_{Z(t)} = \exp\left(\frac{t^2}{2}\right) \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \exp\left(-\frac{(x-t)^2}{2}\right) \mathrm{d}x = \exp\left(\frac{t^2}{2}\right) \tag{22.8}$$

MGF of a General Normal Variable

If

$$X \sim \mathrm{N}(\mu, \sigma^2) \tag{22.9}$$

then

$$M_X(t) = \exp\left(\mu t + \sigma^2 \frac{t^2}{2}\right) \tag{22.10}$$

Proof

Write

$$X = \sigma Z + \mu \tag{22.11}$$

with $Z \sim \mathrm{N}(0, 1)$. Then

$$M_X(t) = \mathbb{E}[\exp(t(\sigma Z + \mu))] = \exp(\mu t)\mathbb{E}[\exp((\sigma t)Z)] = \exp(\mu t)M_{Z(\sigma t)} \tag{22.12}$$

Hence

$$M_X(t) = \exp(\mu t)\exp\left(\sigma^2 \frac{t^2}{2}\right) = \exp\left(\mu t + \sigma^2 \frac{t^2}{2}\right) \tag{22.13}$$

Proof

This also gives another proof of closure under independent sums. If X and Y are independent normals, then

$$M_{X+Y}(t) = M_X(t)M_Y(t) \tag{22.14}$$

$$= \exp\left((\mu + \nu)t + (\sigma^2 + \tau^2)\frac{t^2}{2}\right) \tag{22.15}$$

which is the MGF of

$$\mathrm{N}(\mu + \nu, \sigma^2 + \tau^2) \tag{22.16}$$

7.23 Gaussian Concentration

Theorem

If

$$X \sim \mathrm{N}(\mu, \sigma^2) \tag{23.1}$$

then for every $a > 0$,

$$\Pr(|X - \mu| \geq a\sigma) \leq 2 \exp\left(-\frac{a^2}{2}\right) \tag{23.2}$$

Proof

Standardize

$$Z = \frac{X - \mu}{\sigma} \sim \mathrm{N}(0, 1) \tag{23.3}$$

For any $t > 0$, Markov's inequality gives

$$\Pr(Z \geq a) = \Pr(\exp(tZ) \geq \exp(ta)) \leq \frac{\mathbb{E}[\exp(tZ)]}{\exp(ta)} \tag{23.4}$$

Using the MGF of the standard normal,

$$\Pr(Z \geq a) \leq \exp\left(\frac{t^2}{2} - ta\right) \tag{23.5}$$

The right-hand side is minimized at $t = a$, giving

$$\Pr(Z \geq a) \leq \exp\left(-\frac{a^2}{2}\right) \tag{23.6}$$

By symmetry,

$$\Pr(|Z| \geq a) \leq 2 \exp\left(-\frac{a^2}{2}\right) \tag{23.7}$$

and therefore

$$\Pr(|X - \mu| \geq a\sigma) = \Pr(|Z| \geq a) \leq 2 \exp\left(-\frac{a^2}{2}\right) \tag{23.8}$$

Note

Numerically this is weaker than the empirical 68-95-99.7 rule, but it is much sharper than Chebyshev's inequality and has the advantage of being a clean closed-form bound.

7.24 Bivariate Normal Distribution

Def **standard bivariate normal**

A pair (X, Y) has the **standard bivariate normal distribution** with correlation parameter $-1 < \rho < 1$ if its joint density is

$$f_{X,Y}(x, y) = \frac{1}{2\pi\sqrt{1-\rho^2}} \exp\left(-\frac{x^2 - 2\rho xy + y^2}{2(1-\rho^2)}\right) \tag{24.1}$$

for $x, y \in \mathbb{R}$.

Marginals and Correlation

Under this density,

$$X \sim \mathrm{N}(0, 1), \quad Y \sim \mathrm{N}(0, 1) \tag{24.2}$$

and

$$\mathbf{Cov}(X, Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y] = \mathbb{E}[XY] = \rho \tag{24.3}$$

If $\rho = 0$, then

$$f_{X,Y}(x, y) = \varphi(x)\varphi(y) \tag{24.4}$$

so X and Y are independent.

Note

For bivariate normal variables, being uncorrelated is the same as being independent. This is a special property of the normal family; it is false for general random variables.

Def **general bivariate normal**

A pair (X, Y) has a bivariate normal distribution with means μ_1, μ_2, variances σ_1^2, σ_2^2, and correlation ρ if

$$f_{X,Y}(x, y) = \frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}} \exp\left(-\frac{Q(x, y)}{2}\right) \tag{24.5}$$

where

$$Q(x, y) = \frac{1}{1-\rho^2}\left[\left(\frac{x-\mu_1}{\sigma_1}\right)^2 - 2\rho\left(\frac{x-\mu_1}{\sigma_1}\right)\left(\frac{y-\mu_2}{\sigma_2}\right) + \left(\frac{y-\mu_2}{\sigma_2}\right)^2\right] \tag{24.6}$$

Note

Marginally,

$$X \sim \mathrm{N}(\mu_1, \sigma_1^2), \quad Y \sim \mathrm{N}(\mu_2, \sigma_2^2) \tag{24.7}$$

and

$$\mathbf{Cov}(X, Y) = \sigma_1 \sigma_2 \rho \tag{24.8}$$

7.25 Multivariate Normal Distribution

Def multivariate normal

A random vector $Y = (Y_1, ..., Y_n)$ has a **multivariate normal distribution** if there are independent standard normal variables

$$X_1, ..., X_k \tag{25.1}$$

together with a matrix $A \in \mathbb{R}^{n \times k}$ and a vector $\mu \in \mathbb{R}^n$ such that

$$Y = AX + \mu \tag{25.2}$$

where $X = (X_1, ..., X_k)$.

Density Form

If

$$\Sigma = AA^T \tag{25.3}$$

has full rank, then Y has density

$$f_{Y(y)} = \frac{1}{(2\pi)^{\frac{n}{2}} \sqrt{\det(\Sigma)}} \exp\left(-\frac{1}{2}(y - \mu)^T \Sigma^{-1} (y - \mu)\right) \tag{25.4}$$

We write

$$Y \sim \mathrm{N}(\mu, \Sigma) \tag{25.5}$$

Note

In this notation,

$$Y_i \sim \mathrm{N}(\mu_i, \Sigma_{i,i}), \quad \mathbf{Cov}(Y_i, Y_j) = \Sigma_{i,j} \tag{25.6}$$

An equivalent characterization is that every linear combination

$$a_1 Y_1 + ... + a_n Y_n \tag{25.7}$$

is normally distributed.

7.26 Chi-Squared Distribution

Def chi-squared distribution

If $Z_1, \ldots, Z_k$ are independent standard normal random variables, then

$$Q = \sum_{i=1}^{k} Z_i^2 \tag{26.1}$$

has the **chi-squared distribution** with k degrees of freedom, written

$$Q \sim \chi^2(k) \tag{26.2}$$

Basic Properties

Since

$$\mathbb{E}[Z_i^2] = \mathbf{Var}(Z_i) = 1 \tag{26.3}$$

we have

$$\mathbb{E}[Q] = k \tag{26.4}$$

Also, if $U \sim \chi^2(k)$ and $V \sim \chi^2(l)$ are independent, then

$$U + V \sim \chi^2(k + l) \tag{26.5}$$

Example

Let $Z \sim \mathrm{N}(0, 1)$ and $Y = Z^2$. For $y \geq 0$,

$$F_Y(y) = \Pr(Z^2 \leq y) = \Pr(-\sqrt{y} \leq Z \leq \sqrt{y}) = 2\Phi(\sqrt{y}) - 1 \tag{26.6}$$

Differentiating gives

$$f_Y(y) = \frac{1}{\sqrt{2\pi y}} \exp\left(-\frac{y}{2}\right) \tag{26.7}$$

which is the density of $\chi^2(1)$.

Theorem

For integer $k \geq 1$, the density of $\chi^2(k)$ is

$$f(x) = \frac{1}{2^{\frac{k}{2}}\Gamma\left(\frac{k}{2}\right)} x^{\frac{k}{2}-1} \exp\left(-\frac{x}{2}\right), \quad x \geq 0 \tag{26.8}$$

and $f(x) = 0$ for $x < 0$.

7.27 Gamma Function and Gamma Distribution

Def **gamma function**

The **gamma function** extends the factorial by

$$\Gamma(z) = \int_0^\infty t^{z-1} \exp(-t)\, \mathrm{d}t \tag{27.1}$$

for $z > 0$. In particular,

$$\Gamma(n) = (n-1)! \tag{27.2}$$

for every positive integer n.

Def **gamma distribution**

A random variable X has the **gamma distribution** with shape $k > 0$ and rate $\lambda > 0$, written

$$X \sim \mathrm{Gamma}(k, \lambda) \tag{27.3}$$

if its density is

$$f_X(x) = \frac{\lambda^k}{\Gamma(k)} x^{k-1} \exp(-\lambda x), \quad x \geq 0 \tag{27.4}$$

and $f_X(x) = 0$ for $x < 0$.

Note

The exponential distribution is the special case

$$\mathrm{Gamma}(1, \lambda) = \mathrm{Exp}(\lambda) \tag{27.5}$$

and the chi-squared distribution is

$$\chi^2(k) = \mathrm{Gamma}\left(\frac{k}{2}, \frac{1}{2}\right) \tag{27.6}$$

for integer $k \geq 1$.

MGF of the Gamma Law

If $X \sim \mathrm{Gamma}(k, \lambda)$, then for $t < \lambda$,

$$M_X(t) = \mathbb{E}[\exp(tX)] = \left(1 - \frac{t}{\lambda}\right)^{-k} \tag{27.7}$$

Proof

Directly,

$$M_X(t) = \int_0^\infty \exp(tx) \frac{\lambda^k}{\Gamma(k)} x^{k-1} \exp(-\lambda x)\, \mathrm{d}x \tag{27.8}$$

$$= \frac{\lambda^k}{\Gamma(k)} \int_0^\infty x^{k-1} \exp(-(\lambda - t)x)\, \mathrm{d}x \tag{27.9}$$

Substitute $u = (\lambda - t)x$. Then

$$M_X(t) = \frac{\lambda^k}{(\lambda - t)^k} \cdot \frac{1}{\Gamma(k)} \int_0^\infty u^{k-1} \exp(-u)\, \mathrm{d}u = \left(1 - \frac{t}{\lambda}\right)^{-k} \tag{27.10}$$

Theorem

If $X \sim \mathrm{Gamma}(\alpha, \lambda)$ and $Y \sim \mathrm{Gamma}(\beta, \lambda)$ are independent, then

$$X + Y \sim \mathrm{Gamma}(\alpha + \beta, \lambda) \tag{27.11}$$

Proof

By independence,

$$M_{X+Y}(t) = M_X(t) M_Y(t) \tag{27.12}$$

for $t < \lambda$. Using the gamma MGF,

$$M_{X+Y}(t) = \left(1 - \frac{t}{\lambda}\right)^{-\alpha} \left(1 - \frac{t}{\lambda}\right)^{-\beta} = \left(1 - \frac{t}{\lambda}\right)^{-(\alpha+\beta)} \tag{27.13}$$

This is the MGF of $\mathrm{Gamma}(\alpha + \beta, \lambda)$.

Corollary

If $X_1, \ldots, X_k$ are i.i.d. $\mathrm{Exp}(\lambda)$, then

$$\sum_{i=1}^{k} X_i \sim \mathrm{Gamma}(k, \lambda) \tag{27.14}$$

7.28 Poisson Process from Interarrival Times

Gamma Waiting Times

Let $X_1, X_2, \ldots$ be the interarrival times of a Poisson process with rate λ, so the X_i are i.i.d. $\mathrm{Exp}(\lambda)$. The time of the n-th ring is

$$S_n = \sum_{i=1}^{n} X_i \sim \mathrm{Gamma}(n, \lambda) \tag{28.1}$$

Theorem

For a Poisson process with rate λ,

$$\Pr(N(t) = n) = \exp(-\lambda t) \frac{(\lambda t)^n}{n!} \tag{28.2}$$

for every integer $n \geq 0$.

Proof

For $n = 0$,

$$\Pr(N(t) = 0) = \Pr(X_1 > t) = \exp(-\lambda t) \tag{28.3}$$

For $n \geq 1$, the event $N(t) = n$ means

$$S_n \leq t < S_n + X_{n+1} \tag{28.4}$$

Hence

$$\Pr(N(t) = n) = \int_0^t f_{S_n}(x) \Pr(X_{n+1} > t - x) \, \mathrm{d}x \tag{28.5}$$

Using $S_n \sim \mathrm{Gamma}(n, \lambda)$ and the exponential tail,

$$= \int_0^t \frac{\lambda^n}{(n-1)!} x^{n-1} \exp(-\lambda x) \exp(-\lambda(t - x)) \, \mathrm{d}x \tag{28.6}$$

$$= \exp(-\lambda t) \frac{\lambda^n}{(n-1)!} \int_0^t x^{n-1} \, \mathrm{d}x = \exp(-\lambda t) \frac{(\lambda t)^n}{n!} \tag{28.7}$$

7.29 Beta Distribution

Def beta distribution

A random variable X has the **beta distribution** with parameters $a, b > 0$, written

$$X \sim \mathrm{Beta}(a, b) \tag{29.1}$$

if its density is

$$f_X(x) = \frac{1}{B(a,b)} x^{a-1}(1-x)^{b-1}, \quad 0 \le x \le 1 \tag{29.2}$$

and $f_X(x) = 0$ outside $[0, 1]$.

where

$$B(a,b) = \int_0^1 x^{a-1}(1-x)^{b-1}\,\mathrm{d}x = \frac{\Gamma(a)\Gamma(b)}{\Gamma(a+b)} \tag{29.3}$$

Note

The uniform distribution on $[0, 1]$ is

$$\mathrm{Beta}(1,1) \tag{29.4}$$

If $U_1, \ldots, U_n$ are i.i.d. uniform on $[0, 1]$, then

$$\min(U_1, \ldots, U_n) \sim \mathrm{Beta}(1, n) \tag{29.5}$$

Gamma-Beta Connection

If $X \sim \mathrm{Gamma}(\alpha, \lambda)$ and $Y \sim \mathrm{Gamma}(\beta, \lambda)$ are independent, then

$$\frac{X}{X+Y} \sim \mathrm{Beta}(\alpha, \beta) \tag{29.6}$$

Proof

Use the change of variables

$$R = \frac{X}{X+Y}, \quad S = X + Y \tag{29.7}$$

so that

$$X = RS, \quad Y = (1-R)S \tag{29.8}$$

with Jacobian S. The joint density of (R, S) is proportional to

$$(rs)^{\alpha-1}((1-r)s)^{\beta-1}\exp(-\lambda s)s \tag{29.9}$$

which factors as

$$r^{\alpha-1}(1-r)^{\beta-1}\cdot s^{\alpha+\beta-1}\exp(-\lambda s) \tag{29.10}$$

Thus the marginal density of R is beta with parameters α, β.

7.30 Cauchy Distribution

Def **Cauchy distribution**

A random variable X has the **Cauchy distribution** if its density is

$$f_X(x) = \frac{1}{\pi(1+x^2)}, \quad -\infty < x < \infty \tag{30.1}$$

Note

The Cauchy distribution has heavy tails. For every $k \geq 1$,

$$\mathbb{E}[|X|^k] = \infty \tag{30.2}$$

so the mean, variance, and higher moments are not finite. Its moment generating function does not exist for any nonzero t.

Limit Theorems

设 $X_1, X_2, \cdots$ 是 i.i.d. random variables (独立同分布随机变量)，并且

$$\mu = \mathbb{E}[X_1], \quad \mathbf{Var}[X_1] = \sigma^2 \tag{8.3}$$

记

$$\overline{X}_n = \frac{1}{n}\sum_{i=1}^{n} X_i \tag{8.4}$$

为 **sample mean** (样本均值).

Theorem Law of Large Numbers (LLN), 大数定律

Law of Large Numbers 的核心结论是: sample mean 会收敛到 expectation (期望).

$$\overline{X}_n \to \mu \quad \text{as } n \to \infty \tag{8.5}$$

根据收敛意义不同，它分成两个常见版本：

$$\text{weak LLN: } \overline{X}_n \xrightarrow{P} \mu, \quad \text{strong LLN: } \overline{X}_n \xrightarrow{\text{a.s.}} \mu \tag{8.6}$$

Theorem Central Limit Theorem (CLT), 中心极限定理

Central Limit Theorem 说的是：把 sample mean 去中心化、再按标准差尺度放大后， 极限分布是 standard normal distribution (标准正态分布).

$$Z_n = \frac{\overline{X}_n - \mu}{\frac{\sigma}{\sqrt{n}}} \xrightarrow{D} N(0,1) \quad \text{as } n \to \infty \tag{8.7}$$

8.1 Convergence 收敛

对 ordinary sequence (普通数列) $a_1, a_2, \cdots \in \mathbb{R}$,

$$\begin{aligned} & a_n \to a \\ \Leftrightarrow\ & \forall \varepsilon > 0, \exists N, \text{ s.t. } \forall n \geq N, |a_n - a| < \varepsilon \end{aligned} \tag{1.1}$$

这只是说：给定任意误差尺度 ε, 从某个 N 之后所有项都落在 $(a - \varepsilon, a + \varepsilon)$ 内。

对 function sequence (函数列) $f_1, f_2, \cdots : \Omega \to \mathbb{R}$, **pointwise convergence** (逐点收敛) 到 f 指

$$\forall \omega \in \Omega, \quad \lim_{n \to \infty} f_{n(\omega)} = f(\omega) \tag{1.2}$$

random variable (随机变量) 本身也是函数：

$$X_n : \Omega \to \mathbb{R}, \quad X : \Omega \to \mathbb{R} \tag{1.3}$$

但它还诱导一个 CDF (cumulative distribution function, 分布函数)：

$$F_{X_n} : \mathbb{R} \to [0,1], \quad F_X : \mathbb{R} \to [0,1] \tag{1.4}$$

因此讨论 $X_n \to X$ 时有两条路线：

$$\text{比较样本点上的函数值 } X_{n(\omega)} \quad \text{或比较分布函数 } F_{X_n}(x) \tag{1.5}$$

这正是多种 convergence modes (收敛模式) 的来源。

Example

Coin bits to uniform (硬币二进制展开到均匀分布). 设 $B_1, B_2, \cdots$ 是 i.i.d. $\text{Bern}\left(\frac{1}{2}\right)$, 定义

$$U_n = \sum_{j=1}^{n} \frac{B_j}{2^j}, \quad U = \sum_{j=1}^{\infty} \frac{B_j}{2^j} \tag{1.6}$$

则

$$|U - U_n| \le \sum_{j=n+1}^{\infty} 2^{-j} = 2^{-n}, \tag{1.7}$$

所以 $U_n \xrightarrow{\text{a.s.}} U$.

对任意长度为 2^{-m} 的 dyadic interval (二进区间), 前 m 个 bit 必须等于某个 固定二进制串，概率正好是 2^{-m}. 因此 U 的分布是 $U[0,1]$. 唯一要小心的是 dyadic endpoints (二进端点) 有两种展开，但这些点是 countable set, 概率为 0.

8.1.1 Modes of Convergence

以下都假设 $X, X_1, X_2, \cdots$ 是同一 probability space (概率空间) $(\Omega, \mathcal{F}, \Pr)$ 上的 random variables.

Def convergence in distribution / in law, 依分布收敛

$$\begin{aligned} & X_n \xrightarrow{D} X \\ \Leftrightarrow\ & F_{X_n}(x) \to F_{X(x)} \quad \text{for every continuity point (连续点) } x \text{ of } F_X \end{aligned} \tag{1.8}$$

它也叫 **weak convergence of measures** (测度弱收敛). 重点是：这里只看 distribution (分布)， 不关心 X_n 和 X 在同一个样本点 ω 上是否接近。

Def convergence in probability, 依概率收敛

$$\begin{aligned} & X_n \xrightarrow{P} X \\ \Leftrightarrow\ & \forall \varepsilon > 0, \quad \Pr(|X_n - X| > \varepsilon) \to 0 \end{aligned} \tag{1.9}$$

也就是说：对任意固定误差 ε, “偏差超过 ε” 的概率趋于 0. 从函数角度看，它是 convergence in measure (依测度收敛).

Def almost sure convergence, 几乎必然收敛

$$\begin{aligned} & X_n \xrightarrow{\text{a.s.}} X \\ \Leftrightarrow\ & \exists A \in \mathcal{F}, \Pr(A) = 1, \text{and } \forall \omega \in A, \lim_{n\to\infty} X_{n(\omega)} = X(\omega) \end{aligned} \tag{1.10}$$

等价地，

$$\Pr(\lim_{n\to\infty} X_n = X) = 1 \tag{1.11}$$

它是“除去一个 probability 0 的坏集合后，逐点收敛”。

Three modes, 三种模式

$$X_n \xrightarrow{D} X : \text{分布形状收敛} \quad F_{X_n} \to F_X \tag{1.12}$$

$$X_n \xrightarrow{P} X : \text{绝大多数样本点上接近} \quad \Pr(|X_n - X| > \varepsilon) \to 0 \tag{1.13}$$

$$X_n \xrightarrow{\text{a.s.}} X : \text{几乎每条样本路径最终都接近} \tag{1.14}$$

8.1.2 Convergence in Distribution

definition 中只要求在 F_X 的 continuity points (连续点) 上收敛，这是必要的。考虑 $X_n \sim U\left(0, \frac{1}{n}\right)$, 令 $X = 0$ 以概率 1 成立。则

$$F_{X_n}(x) = \begin{cases} 0 & x \leq 0 \\ nx & 0 < x < \frac{1}{n} \\ 1 & x \geq \frac{1}{n} \end{cases} \tag{1.15}$$

而

$$F_{X(x)} = \begin{cases} 0 & x < 0 \\ 1 & x \geq 0. \end{cases} \tag{1.16}$$

对每个 $x \neq 0$, 都有 $F_{X_n}(x) \to F_{X(x)}$, 因而 $X_n \xrightarrow{D} X$. 但在 $x = 0$ 处，

$$F_{X_n}(0) = 0 \quad \text{while} \quad F_{X(0)} = 1 \tag{1.17}$$

所以如果强行要求所有点都收敛，依分布收敛就会被错误排除。

Distribution only

convergence in distribution 只依赖 distribution:

$$X_n \xrightarrow{D} X \;\text{ 且 }\; F_X = F_Y \quad \Rightarrow \quad X_n \xrightarrow{D} Y \tag{1.18}$$

换句话说，只要 X 和 Y 同分布，它们作为极限在 $\xrightarrow{D}$ 下没有区别。

Example

依分布收敛不推出依概率收敛。设 $X \sim U[0,1]$, 且对所有 n 令

$$X_n = 1 - X \tag{1.19}$$

因为 $1 - X \sim U[0,1]$, 所以

$$F_{X_n} = F_X \quad \text{for all } n, \tag{1.20}$$

从而 $X_n \xrightarrow{D} X$. 但是

$$\Pr\left(|X_n - X| > \frac{1}{4}\right) = \Pr\left(|1 - 2X| > \frac{1}{4}\right) = \frac{3}{4} \tag{1.21}$$

对所有 n 都不变，因此不可能 $X_n \xrightarrow{P} X$.

8.1.3 Convergence in Probability

Theorem $X_n \xrightarrow{P} X$ implies $X_n \xrightarrow{D} X$

convergence in probability 比 convergence in distribution 更强：

$$X_n \xrightarrow{P} X \quad \Rightarrow \quad X_n \xrightarrow{D} X \tag{1.22}$$

Proof

固定 $\varepsilon > 0$. 对任意 $x \in \mathbb{R}$,

$$\{X_n \leq x\} \subseteq \{X \leq x + \varepsilon\} \cup \{|X_n - X| > \varepsilon\} \tag{1.23}$$

原因是：若 $X_n \leq x$ 且 $|X_n - X| \leq \varepsilon$, 则 $X \leq x + \varepsilon$. 因此

$$F_{X_n}(x) \leq F_{X(x+\varepsilon)} + \Pr(|X_n - X| > \varepsilon) \tag{1.24}$$

同理，

$$\{X \leq x - \varepsilon\} \subseteq \{X_n \leq x\} \cup \{|X_n - X| > \varepsilon\}, \tag{1.25}$$

所以

$$F_{X(x-\varepsilon)} - \Pr(|X_n - X| > \varepsilon) \leq F_{X_n}(x) \tag{1.26}$$

令 $n \to \infty$, 由于 $X_n \xrightarrow{P} X$,

$$F_{X(x-\varepsilon)} \leq \liminf_{n\to\infty} F_{X_n}(x) \leq \limsup_{n\to\infty} F_{X_n}(x) \leq F_{X(x+\varepsilon)} \tag{1.27}$$

如果 F_X 在 x 连续，再令 $\varepsilon \to 0^+$, 得

$$F_{X(x-\varepsilon)} \to F_{X(x)}, \quad F_{X(x+\varepsilon)} \to F_{X(x)} \tag{1.28}$$

夹逼即得 $F_{X_n}(x) \to F_{X(x)}$.

Theorem Convergence to a constant, 常数极限

如果极限是常数 $c \in \mathbb{R}$, 那么

$$X_n \xrightarrow{D} c \quad \Rightarrow \quad X_n \xrightarrow{P} c \tag{1.29}$$

Proof

常数 c 的 CDF 是

$$F_{c(x)} = \begin{cases} 0 & x < c \\ 1 & x \geq c. \end{cases} \tag{1.30}$$

对任意 $\varepsilon > 0$, $c - \varepsilon$ 和 $c + \varepsilon$ 都是 F_c 的连续点， 所以

$$F_{X_n}(c - \varepsilon) \to 0, \quad F_{X_n}(c + \varepsilon) \to 1 \tag{1.31}$$

因而

$$\Pr(|X_n - c| > \varepsilon) \leq \Pr(X_n \leq c - \varepsilon) + \Pr(X_n > c + \varepsilon) \tag{1.32}$$

$$= F_{X_n}(c - \varepsilon) + 1 - F_{X_n}(c + \varepsilon) \to 0 \tag{1.33}$$

这正是 $X_n \xrightarrow{P} c$.

8.1.4 Almost Sure Convergence

事件“X_n 收敛到 X”可以写成 countable set operations (可数集合运算):

$$\{\lim_{n\to\infty} X_n = X\} = \cap_{m=1}^{\infty} \cup_{n_0=1}^{\infty} \cap_{n=n_0}^{\infty} \left\{|X_n - X| \leq \frac{1}{m}\right\} \tag{1.34}$$

含义是：对每个精度 $\frac{1}{m}$, 都存在某个时刻 n_0, 之后所有 X_n 都在这个精度内。

8.2 Borel-Cantelli Lemmas

设 $A_1, A_2, \cdots$ 是一列 events (事件). 定义

$$\{A_n \quad \text{infinitely often}\} = \cap_{n=1}^{\infty} \cup_{m=n}^{\infty} A_m \tag{2.1}$$

记作 A_n i.o. (infinitely often, 无穷多次发生).

Theorem Borel-Cantelli I, 第一引理

如果

$$\sum_{n=1}^{\infty} \Pr(A_n) < \infty, \tag{2.2}$$

那么

$$\Pr(A_n \quad \text{i.o.}) = 0 \tag{2.3}$$

Proof

由 union bound (并集界),

$$\Pr(\cup_{m=n}^{\infty} A_m) \leq \sum_{m=n}^{\infty} \Pr(A_m) \tag{2.4}$$

若 $\sum_n \Pr(A_n)$ 收敛，则 tail sum (尾和) 满足

$$\sum_{m=n}^{\infty} \Pr(A_m) \to 0 \tag{2.5}$$

令 $B_n = \cup_{m=n}^{\infty} A_m$. 这是一列 decreasing events (递减事件), 且

$$\cap_{n=1}^{\infty} B_n = \{A_n \quad \text{i.o.}\} \tag{2.6}$$

由 continuity from above,

$$\Pr(A_n \quad \text{i.o.}) = \lim_{n\to\infty} \Pr(B_n) \leq \lim_{n\to\infty} \sum_{m=n}^{\infty} \Pr(A_m) = 0 \tag{2.7}$$

Theorem Borel-Cantelli II, 第二引理

如果 $A_1, A_2, \cdots$ independent (相互独立), 且

$$\sum_{n=1}^{\infty} \Pr(A_n) = \infty, \tag{2.8}$$

那么

$$\Pr(A_n \quad \text{i.o.}) = 1 \tag{2.9}$$

Proof

固定 n 和 $N > n$. 由 independence,

$$\Pr(\cap_{m=n}^{N} A_m^c) = \prod_{m=n}^{N} (1 - \Pr(A_m)) \leq \exp\left(-\sum_{m=n}^{N} \Pr(A_m)\right) \tag{2.10}$$

这里用到 $1 - u \leq e^{-u}$. 因为级数发散，令 $N \to \infty$ 得

$$\Pr(\cap_{m=n}^{\infty} A_m^c) = 0 \tag{2.11}$$

又

$$\{A_n \quad \text{i.o.}\}^c = \cup_{n=1}^{\infty} \cap_{m=n}^{\infty} A_m^c \tag{2.12}$$

是 countable union (可数并) 的 null events (零概率事件), 所以概率仍为 0. 因而 $\Pr(A_n \quad \text{i.o.}) = 1$.

Continuity of probability measures, 概率测度连续性

若 $B_1 \subseteq B_2 \subseteq \cdots$ 且 $B = \cup_{n=1}^{\infty} B_n$, 则

$$\Pr(B) = \lim_{n\to\infty} \Pr(B_n) \tag{2.13}$$

证明思路是把 B 写成 disjoint union (不交并):

$$B = B_1 \cup (B_2 - B_1) \cup (B_3 - B_2) \cup \cdots \tag{2.14}$$

然后概率求和 telescope (望远镜相消).

若 $C_1 \supseteq C_2 \supseteq \cdots$ 且 $C = \cap_{n=1}^{\infty} C_n$, 则

$$\Pr(C) = \lim_{n\to\infty} \Pr(C_n) \tag{2.15}$$

这是对 complements (补集) 使用递增情形得到的。

Theorem Lemma: a.s. convergence implies convergence in probability

almost sure convergence 比 convergence in probability 更强:

$$X_n \xrightarrow{\text{a.s.}} X \quad \Rightarrow \quad X_n \xrightarrow{P} X \tag{2.16}$$

Proof

固定任意 $\varepsilon > 0$, 令

$$A_{n(\varepsilon)} = \left\{\omega \in \Omega : \left|X_{n(\omega)} - X(\omega)\right| > \varepsilon\right\} \tag{2.17}$$

这里 $A_{n(\varepsilon)}$ 是第 n 步偏离超过 ε 的 bad event (坏事件).

若 $X_{n(\omega)} \to X(\omega)$, 这里的 $\to$ 是 ordinary pointwise convergence (普通逐点收敛), 则对这个 ε 存在 $n_0 = n_0(\omega, \varepsilon)$, 使得所有 $m \geq n_0$ 都满足 $\omega \in A^c_{m(\varepsilon)}$. 因此

$$\left\{\omega \in \Omega : \lim_{n\to\infty} X_{n(\omega)} = X(\omega)\right\} \subseteq \cup_{n=1}^{\infty} \cap_{m=n}^{\infty} A^c_{m(\varepsilon)} \tag{2.18}$$

假设 $X_n \xrightarrow{\text{a.s.}} X$, 即

$$\Pr(\lim_{n\to\infty} X_n = X) = 1 \tag{2.19}$$

由上面的包含关系,

$$1 = \Pr(\lim_{n\to\infty} X_n = X) \leq \Pr\left(\cup_{n=1}^{\infty} \cap_{m=n}^{\infty} A^c_{m(\varepsilon)}\right) \leq 1 \tag{2.20}$$

所以

$$\Pr\left(\cup_{n=1}^{\infty} \cap_{m=n}^{\infty} A_{m(\varepsilon)}^{c}\right) = 1 \tag{2.21}$$

取 complement (补集), 得

$$0 = \Pr\left(\cap_{n=1}^{\infty} \cup_{m=n}^{\infty} A_{m(\varepsilon)}\right) \tag{2.22}$$

令

$$B_{n(\varepsilon)} = \cup_{m=n}^{\infty} A_{m(\varepsilon)} \tag{2.23}$$

则 $B_1(\varepsilon) \supseteq B_2(\varepsilon) \supseteq \cdots$. 由上面的 probability continuity from above (概率测度上连续性),

$$0 = \Pr\left(\cap_{n=1}^{\infty} B_{n(\varepsilon)}\right) = \lim_{n\to\infty} \Pr\left(B_{n(\varepsilon)}\right) = \lim_{n\to\infty} \Pr\left(\cup_{m=n}^{\infty} A_{m(\varepsilon)}\right) \tag{2.24}$$

最后，

$$\Pr(|X_n - X| > \varepsilon) = \Pr\left(A_{n(\varepsilon)}\right) \le \Pr\left(\cup_{m=n}^{\infty} A_{m(\varepsilon)}\right) \to 0 \tag{2.25}$$

因为 $\varepsilon > 0$ 任意，所以 $X_n \xrightarrow{P} X$.

Theorem A useful condition for a.s. convergence, 几乎必然收敛判别

如果

$$\forall \varepsilon > 0, \quad \sum_{n=1}^{\infty} \Pr(|X_n - X| > \varepsilon) < \infty, \tag{2.26}$$

那么

$$X_n \xrightarrow{\text{a.s.}} X \tag{2.27}$$

Proof

对每个整数 $k \ge 1$, 把 Borel-Cantelli I 用在

$$A_{n(\frac{1}{k})} = \left\{|X_n - X| > \frac{1}{k}\right\} \tag{2.28}$$

上，得到

$$\Pr\left(A_{n(\frac{1}{k})} \;\; \text{i.o.}\right) = 0 \tag{2.29}$$

对 k 再做 countable union, 概率仍为 0. 在这个零概率坏集合之外， 对每个 k 都存在 N_k, 使得

$$n \geq N_k \Rightarrow |X_n - X| \leq \frac{1}{k} \tag{2.30}$$

这就等价于 $X_{n(\omega)} \to X(\omega)$.

Example

In probability does not imply a.s. (依概率不推出几乎必然). 设 $X_n \sim \mathrm{Bern}(\frac{1}{n})$ 且相互独立。对任意 $\varepsilon \in (0,1)$,

$$\Pr(|X_n - 0| > \varepsilon) = \Pr(X_n = 1) = \frac{1}{n} \to 0, \tag{2.31}$$

所以 $X_n \xrightarrow{P} 0$. 但是

$$\sum_{n=1}^{\infty} \Pr(X_n = 1) = \sum_{n=1}^{\infty} \frac{1}{n} = \infty \tag{2.32}$$

由 Borel-Cantelli II,

$$\Pr(X_n = 1 \quad \text{i.o.}) = 1 \tag{2.33}$$

因此几乎每条样本路径上都会出现无穷多个 1, 不可能几乎必然收敛到 0.

8.3 Strength of Convergence

Main implication chain, 强弱关系

$$X_n \xrightarrow{\text{a.s.}} X \quad \Rightarrow \quad X_n \xrightarrow{P} X \quad \Rightarrow \quad X_n \xrightarrow{D} X \tag{3.1}$$

一般来说两个箭头都不能反过来。唯一常用例外是 constant limit (常数极限):

$$X_n \xrightarrow{D} c \quad \Rightarrow \quad X_n \xrightarrow{P} c \tag{3.2}$$

8.3.1 Coupling and Continuous Mapping

Theorem Skorokhod representation, 斯科罗霍德表示

若 $X_n \xrightarrow{D} X$, 则可以在另一个 probability space 上构造 $Y_1, Y_2, \cdots, Y$, 使得

$$F_{Y_n} = F_{X_n}, \quad F_Y = F_X, \quad \text{and} \quad Y_n \xrightarrow{\text{a.s.}} Y \tag{3.3}$$

Proof

用 inverse transform sampling (反函数抽样). 取新空间为 $([0,1],\mathcal{B},\mathrm{Leb})$, 对 $u\in[0,1]$ 定义

$$Y_{n(u)}=\inf\{x\in\mathbb{R}:u\le F_{X_n}(x)\},\quad Y(u)=\inf\{x\in\mathbb{R}:u\le F_{X(x)}\}\tag{3.4}$$

则 Y_n 与 X_n 同分布，Y 与 X 同分布。由 $F_{X_n}\to F_X$ 在连续点上的收敛，可以验证 $Y_{n(u)}\to Y(u)$ 在 Y 的 quantile map (分位函数) 连续点成立。不连续点集合至多可数或为 null set, 因而得到 almost sure convergence.

Theorem Continuous Mapping Theorem, 连续映射定理

若 $g:\mathbb{R}\to\mathbb{R}$ continuous (连续), 则

$$X_n\xrightarrow{D}X\Rightarrow g(X_n)\xrightarrow{D}g(X)\tag{3.5}$$

$$X_n\xrightarrow{P}X\Rightarrow g(X_n)\xrightarrow{P}g(X)\tag{3.6}$$

$$X_n\xrightarrow{\text{a.s.}}X\Rightarrow g(X_n)\xrightarrow{\text{a.s.}}g(X)\tag{3.7}$$

Proof

a.s. 情形最直接：若 $X_{n(\omega)}\to X(\omega)$ 且 g 连续，则

$$g\left(X_{n(\omega)}\right)\to g(X(\omega))\tag{3.8}$$

probability 情形可以理解为同一个思路加上 compact control (紧集控制)：固定 M, 在 $[-M,M]$ 上连续函数一致连续，所以存在 δ, 使 $|x-y|<\delta$ 推出 $|g(x)-g(y)|<\varepsilon$. 再把事件分成 “X 落在 $[-M,M]$ 内” 与 “X_n 和 X 足够接近” 两部分控制，最后令 $M\to\infty$.

distribution 情形用 Skorokhod representation：选取同分布的 Y_n,Y 使 $Y_n\xrightarrow{\text{a.s.}}Y$. 由 a.s. 情形得到 $g(Y_n)\xrightarrow{\text{a.s.}}g(Y)$, 于是 $g(Y_n)\xrightarrow{D}g(Y)$. 因为分布一致，结论转回 $g(X_n)$.

8.3.2 Other Convergence Modes

Def convergence in mean and L^r, 均值收敛

$X_n\xrightarrow{1}X$ 表示 convergence in mean (均值收敛):

$$\mathbb{E}[|X_n-X|]\to 0\tag{3.9}$$

更一般地，对 $r\ge 1$, $X_n\xrightarrow{r}X$ 表示 convergence in L^r (按 r 阶均值收敛):

$$\mathbb{E}[|X_n-X|^r]\to 0\tag{3.10}$$

L^r **implications,** L^r 收敛的推出关系

Markov's inequality (马尔可夫不等式) 给出

$$\Pr(|X_n - X| > \varepsilon) \le \frac{\mathbb{E}[|X_n - X|^r]}{\varepsilon^r} \tag{3.11}$$

因此

$$X_n \xrightarrow{r} X \Rightarrow X_n \xrightarrow{P} X \tag{3.12}$$

若 $s \ge r \ge 1$, 则由 Jensen/Hölder inequality,

$$\mathbb{E}[|X_n - X|^r] \le \mathbb{E}[|X_n - X|^s]^{\frac{r}{s}} \tag{3.13}$$

所以

$$X_n \xrightarrow{s} X \Rightarrow X_n \xrightarrow{r} X \tag{3.14}$$

8.4 Characteristic Functions

moment generating function (MGF, 矩母函数) 定义为

$$M_{X(t)} = \mathbb{E}[e^{tX}] \tag{4.1}$$

但它未必在 $t = 0$ 附近有限。

characteristic function (特征函数) 定义为

$$\varphi_{X(t)} = \mathbb{E}[e^{itX}], \quad i^2 = -1 \tag{4.2}$$

它总是存在，因为 $|e^{itX}| = 1$.

Fourier viewpoint, Fourier 视角

$$\varphi_{X(t)} = \int_{-\infty}^{\infty} e^{itx}\, \mathrm{d}F_{X(x)} = \mathbb{E}[\cos(tX)] + i\mathbb{E}[\sin(tX)] \tag{4.3}$$

如果 X 有 density (密度) f_X, 则

$$\varphi_{X(t)} = \int_{-\infty}^{\infty} e^{itx} f_{X(x)}\, \mathrm{d}x \tag{4.4}$$

也就是说 characteristic function 是 distribution 的 Fourier transform (傅里叶变换).

Theorem Basic bounds and Taylor expansion, 基本界与展开

对任意 random variable X,

$$\left|\varphi_{X(t)}\right| \leq 1 \tag{4.5}$$

如果 $\mathbb{E}[|X|^k] < \infty$, 则当 $t \to 0$ 时

$$\varphi_{X(t)} = \sum_{j=0}^{k} \left(\frac{\mathbb{E}[X^j]}{j!}\right)(it)^j + o\left(t^k\right) \tag{4.6}$$

特别地，如果 $\mathbb{E}[|X|] < \infty$,

$$\varphi_{X(t)} = 1 + i\mu t + o(t) \tag{4.7}$$

如果 $\mathbb{E}[X^2] < \infty$,

$$\varphi_{X(t)} = 1 + i\mu t - \left(\frac{\mathbb{E}[X^2]}{2}\right)t^2 + o\left(t^2\right) \tag{4.8}$$

Proof

boundedness 来自 triangle inequality:

$$\left|\varphi_{X(t)}\right| = |\mathbb{E}[e^{itX}]| \leq \mathbb{E}\left[\left|e^{itX}\right|\right] = 1 \tag{4.9}$$

对 expansion (展开), 使用 Taylor formula (泰勒公式):

$$e^{itX} = \sum_{j=0}^{k} \frac{(itX)^j}{j!} + R_{k(tX)} \tag{4.10}$$

其中当 $u \to 0$ 时 $R_{k(u)} = o\left(|u|^k\right)$. 若 $\mathbb{E}[|X|^k] < \infty$, dominated convergence (控制收敛) 允许对余项取期望:

$$\mathbb{E}\left[R_{k(tX)}\right] = o\left(t^k\right) \tag{4.11}$$

Standardized expansion, 标准化后的展开

若

$$Y = \frac{X - \mu}{\sigma}, \quad \mathbb{E}[Y] = 0, \quad \mathbb{E}[Y^2] = 1, \tag{4.12}$$

则

$$\varphi_{Y(t)} = 1 - \frac{t^2}{2} + o\left(t^2\right) \tag{4.13}$$

这正是 CLT 证明中每个 summand (求和项) 需要的局部展开。

8.4.1 Normal Characteristic Function

Theorem Normal characteristic function, 正态特征函数

若 $Z \sim N(0,1)$, 则

$$\varphi_{Z(t)} = \mathbb{E}[e^{itZ}] = e^{-\frac{t^2}{2}} \tag{4.14}$$

Proof

因为 Z 的 density 是 even function (偶函数), imaginary part 为 0:

$$\varphi_{Z(t)} = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \cos(tx) e^{-\frac{x^2}{2}}\, \mathrm{d}x \tag{4.15}$$

对 t 求导:

$$\varphi'_{Z(t)} = -\frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} x \sin(tx) e^{-\frac{x^2}{2}}\, \mathrm{d}x \tag{4.16}$$

又因为 $\mathrm{d}\left(e^{-\frac{x^2}{2}}\right) = -xe^{-\frac{x^2}{2}}\, \mathrm{d}x$, integration by parts (分部积分) 给出

$$\int_{-\infty}^{\infty} x \sin(tx) e^{-\frac{x^2}{2}}\, \mathrm{d}x = t \int_{-\infty}^{\infty} \cos(tx) e^{-\frac{x^2}{2}}\, \mathrm{d}x \tag{4.17}$$

因此

$$\varphi'_{Z(t)} = -t\varphi_{Z(t)} \tag{4.18}$$

解这个 ODE (常微分方程), 并用初值 $\varphi_{Z(0)} = 1$, 得

$$\varphi_{Z(t)} = e^{-\frac{t^2}{2}} \tag{4.19}$$

Linear transformation and sums, 线性变换与独立和

如果 $Y = aX + b$, 则

$$\varphi_{Y(t)} = \mathbb{E}\left[e^{it(aX+b)}\right] = e^{itb}\varphi_{X(at)} \tag{4.20}$$

如果 X 和 Y independent, 则

$$\varphi_{X+Y}(t) = \mathbb{E}[e^{itX}e^{itY}] = \mathbb{E}[e^{itX}]\mathbb{E}[e^{itY}] = \varphi_{X(t)}\varphi_{Y(t)} \tag{4.21}$$

Theorem Lévy Continuity Theorem, 连续性定理

对 random variables X_n 和 X,

$$X_n \xrightarrow{D} X \quad \Leftrightarrow \quad \forall t \in \mathbb{R}, \varphi_{X_n}(t) \to \varphi_{X(t)} \tag{4.22}$$

Note

更一般地，如果 $\varphi_{X_n}(t)$ pointwise converges (逐点收敛) 到某个 $\varphi(t)$, 且 φ 在 0 连续，那么 φ 是某个 distribution 的 characteristic function, 并且 X_n 依分布收敛到该 distribution.

Example

Convolution of normal distributions (正态分布卷积). 若

$$X \sim N(\mu, \sigma^2), \quad Y \sim N(\nu, \tau^2) \tag{4.23}$$

independent, 则

$$X + Y \sim N(\mu + \nu, \sigma^2 + \tau^2) \tag{4.24}$$

证明只需要 characteristic function. 设 $Z \sim N(0, 1)$, 则

$$\varphi_{X(t)} = e^{it\mu}\varphi_{Z(\sigma t)} = e^{it\mu - \sigma^2 \frac{t^2}{2}} \tag{4.25}$$

同理

$$\varphi_{Y(t)} = e^{it\nu - \tau^2 \frac{t^2}{2}} \tag{4.26}$$

由 independence,

$$\varphi_{X+Y}(t) = \varphi_{X(t)}\varphi_{Y(t)} = e^{it(\mu+\nu) - (\sigma^2+\tau^2)\frac{t^2}{2}}, \tag{4.27}$$

这正是 $N(\mu + \nu, \sigma^2 + \tau^2)$ 的 characteristic function.

8.5 Law of Large Numbers

8.5.1 Bernoulli's Law of Large Numbers

Theorem Bernoulli LLN, 伯努利大数定律

设 $X_1, X_2, \cdots$ 是 i.i.d. $\text{Bern}(p)$, 并令

$$\overline{X}_n = \frac{X_1 + \cdots + X_n}{n} \tag{5.1}$$

则对任意 $\varepsilon > 0$,

$$\Pr\left(\left|\overline{X}_n - p\right| > \varepsilon\right) \to 0 \tag{5.2}$$

即 $\overline{X}_n \xrightarrow{P} p$.

Proof

因为 $\mathbb{E}[X_i] = p$ 且 $\mathbf{Var}[X_i] = p(1-p)$,

$$\mathbb{E}\left[\overline{X}_n\right] = p, \quad \mathbf{Var}\left[\overline{X}_n\right] = \frac{1}{n^2}\sum_{i=1}^{n}\mathbf{Var}[X_i] = \frac{p(1-p)}{n} \tag{5.3}$$

由 Chebyshev's inequality (切比雪夫不等式),

$$\Pr\left(\left|\overline{X}_n - p\right| > \varepsilon\right) \le \frac{\mathbf{Var}\left[\overline{X}_n\right]}{\varepsilon^2} = \frac{p(1-p)}{n\varepsilon^2} \to 0 \tag{5.4}$$

8.5.2 Weak LLN Assuming Bounded Variance

Theorem Weak LLN with bounded variance, 有界方差弱大数

设 $X_1, X_2, \cdots$ independent, 且

$$\mathbb{E}[X_i] = \mu, \quad \mathbf{Var}[X_i] \le \sigma^2 < \infty \tag{5.5}$$

令 $\overline{X}_n = \frac{1}{n}\sum_{i=1}^{n} X_i$. 则

$$\overline{X}_n \xrightarrow{P} \mu \tag{5.6}$$

Proof

先算 expectation:

$$\mathbb{E}\left[\overline{X}_n\right] = \frac{1}{n}\sum_{i=1}^{n}\mathbb{E}[X_i] = \mu \tag{5.7}$$

independence 使 covariance terms 消失:

$$\mathbf{Var}\left[\overline{X}_n\right] = \frac{1}{n^2}\sum_{i=1}^{n}\mathbf{Var}[X_i] \le \frac{\sigma^2}{n} \tag{5.8}$$

因此由 Chebyshev,

$$
\Pr\left(\left|\overline{X}_n - \mu\right| > \varepsilon\right) \leq \frac{\mathbf{Var}\left[\overline{X}_n\right]}{\varepsilon^2} \leq \frac{\sigma^2}{n\varepsilon^2} \to 0 \tag{5.9}
$$

8.5.3 Khinchin Weak LLN

Theorem Khinchin's Weak LLN, 辛钦弱大数定律

设 $X_1, X_2, \cdots$ i.i.d., 且只有 finite mean (有限均值) $\mathbb{E}[X_1] = \mu$. 不要求 finite variance. 则

$$
\overline{X}_n \xrightarrow{P} \mu \tag{5.10}
$$

Proof

因为 $\mathbb{E}[|X_1|] < \infty$, characteristic function 有一阶展开:

$$
\varphi_{X(t)} = 1 + i\mu t + o(t) \quad \text{as } t \to 0 \tag{5.11}
$$

对 sample mean,

$$
\varphi_{\overline{X}_n}(t) = \varphi_{X_1 + \cdots + X_n}\left(\frac{t}{n}\right) = \prod_{j=1}^{n} \varphi_{X\left(\frac{t}{n}\right)} \tag{5.12}
$$

这里使用了 independence 和 identical distribution. 于是

$$
\varphi_{\overline{X}_n}(t) = \left(1 + i\mu \frac{t}{n} + o\left(\frac{1}{n}\right)\right)^n \to e^{i\mu t} \tag{5.13}
$$

右边是常数 random variable μ 的 characteristic function. 由 Lévy theorem,

$$
\overline{X}_n \xrightarrow{D} \mu \tag{5.14}
$$

又因为极限是常数，所以

$$
\overline{X}_n \xrightarrow{P} \mu \tag{5.15}
$$

8.5.4 Strong LLN

Theorem Kolmogorov Strong LLN, 柯尔莫哥洛夫强大数定律

设 $X_1, X_2, \cdots$ i.i.d., $\mathbb{E}[|X_1|] < \infty$, 且 $\mathbb{E}[X_1] = \mu$. 则

$$
\overline{X}_n \xrightarrow{\text{a.s.}} \mu \tag{5.16}
$$

Note

完整证明比 weak LLN 深：通常要 truncate (截断) 大值，用 maximal inequality (最大不等式) 控制截断后的中心化和，再配合 Borel-Cantelli.

直观差别是：

$$\text{weak LLN: 每个固定的大 } n \text{ 上偏差概率很小} \tag{5.17}$$

$$\text{strong LLN: 几乎每条样本路径上只发生有限次大偏差} \tag{5.18}$$

A simple strong-law proof under a fourth moment, 四阶矩条件下的简单证明

额外假设 $\mathbb{E}\left[(X_1-\mu)^4\right] < \infty$. 令

$$Y_i = X_i - \mu, \quad S_n = \sum_{i=1}^{n} Y_i \tag{5.19}$$

则 $\mathbb{E}[Y_i] = 0$ 且

$$\overline{X}_n - \mu = \frac{S_n}{n} \tag{5.20}$$

展开 S_n^4 时，任何含有未配对 mean-zero 因子的项，期望都为 0. 因此只剩：

$$\mathbb{E}[S_n^4] = n\mathbb{E}[Y_1^4] + 6\binom{n}{2}\mathbb{E}[Y_1^2]^2 = O(n^2) \tag{5.21}$$

由 Markov inequality,

$$\Pr\left(|\overline{X}_n - \mu| > \varepsilon\right) = \Pr(|S_n| > n\varepsilon) \le \frac{\mathbb{E}[S_n^4]}{n^4\varepsilon^4} = O\left(\frac{1}{n^2}\right) \tag{5.22}$$

所以对每个 $\varepsilon > 0$,

$$\sum_{n=1}^{\infty} \Pr\left(|\overline{X}_n - \mu| > \varepsilon\right) < \infty \tag{5.23}$$

由前面的 a.s. convergence 判别法，得到

$$\overline{X}_n \xrightarrow{\text{a.s.}} \mu \tag{5.24}$$

8.6 De Moivre-Laplace Theorem

Theorem De Moivre-Laplace, 棣莫弗-拉普拉斯定理

设 $p \in (0,1)$, $q = 1-p$, 且 $X_n \sim \mathrm{Bin}(n,p)$. 则

$$\frac{X_n - np}{\sqrt{npq}} \xrightarrow{D} N(0,1) \tag{6.1}$$

Local normal approximation, 局部正态近似

对任意固定 $r > 0$, 当整数 k 满足

$$\frac{|k - np|}{\sqrt{npq}} \leq r \tag{6.2}$$

时，有 uniform approximation (一致近似)

$$\binom{n}{k} p^k q^{n-k} = (1 + o(1)) \frac{1}{\sqrt{2\pi npq}} \exp\left(-\frac{(k-np)^2}{2npq}\right) \tag{6.3}$$

Proof

写

$$k = np + x\sqrt{npq} \tag{6.4}$$

其中 x bounded. 用 Stirling formula (斯特林公式):

$$\binom{n}{k} p^k q^{n-k} = \frac{n^n}{k^k (n-k)^{n-k}} p^k q^{n-k} \cdot \sqrt{\frac{n}{2\pi k(n-k)}} (1 + o(1)) \tag{6.5}$$

取 logarithm (对数), 并代入

$$\frac{k}{n} = p + x\sqrt{p\frac{q}{n}} \tag{6.6}$$

在最大点 $k = np$ 附近，一阶项抵消，二阶项给出

$$-\frac{x^2}{2} = -\frac{(k-np)^2}{2npq} \tag{6.7}$$

前面的 square-root prefactor (平方根因子) 收敛到 $\frac{1}{\sqrt{2\pi npq}}$. 这得到局部近似；再对标准化区间内的 k 求和， 就得到 distributional convergence 到 $N(0,1)$.

8.7 Central Limit Theorem

Theorem Lindeberg-Levy CLT, 林德伯格-列维中心极限定理

设 $X_1, X_2, \cdots$ i.i.d., 且

$$\mathbb{E}[X_1] = \mu, \quad \mathbf{Var}[X_1] = \sigma^2 \in (0, \infty) \tag{7.1}$$

令

$$Z_n = \frac{\overline{X}_n - \mu}{\frac{\sigma}{\sqrt{n}}} = \frac{1}{\sqrt{n}} \sum_{j=1}^{n} \frac{X_j - \mu}{\sigma} \tag{7.2}$$

则

$$Z_n \xrightarrow{D} Z, \quad Z \sim N(0,1) \tag{7.3}$$

Proof

先把每个 summand (求和项) 标准化:

$$Y_j = \frac{X_j - \mu}{\sigma}, \quad \mathbb{E}\left[Y_j\right] = 0, \quad \mathbb{E}\left[Y_j^2\right] = 1 \tag{7.4}$$

则

$$Z_n = \frac{Y_1 + \cdots + Y_n}{\sqrt{n}} \tag{7.5}$$

Y_j 的 characteristic function 满足

$$\varphi_{Y(t)} = 1 - \frac{t^2}{2} + o(t^2) \quad \text{as } t \to 0 \tag{7.6}$$

由 independence,

$$\varphi_{Z_n}(t) = \varphi_{Y_1 + \cdots + Y_n}\left(\frac{t}{\sqrt{n}}\right) = \prod_{j=1}^{n} \varphi_{Y\left(\frac{t}{\sqrt{n}}\right)} \tag{7.7}$$

因为所有因子相同,

$$\varphi_{Z_n}(t) = \left(1 - \frac{t^2}{2n} + o\left(\frac{1}{n}\right)\right)^n \to e^{-\frac{t^2}{2}} \tag{7.8}$$

而 $e^{-\frac{t^2}{2}}$ 正是 $N(0,1)$ 的 characteristic function. 由 Lévy continuity theorem,

$$Z_n \xrightarrow{D} N(0,1) \tag{7.9}$$

Reading the normalization, 为什么这样标准化

总和的 expectation 和 variance 是

$$\mathbb{E}\left[\sum_{i=1}^{n} X_i\right] = n\mu, \quad \mathbf{Var}\left[\sum_{i=1}^{n} X_i\right] = n\sigma^2 \tag{7.10}$$

因此自然的 centered and unit-variance variable (中心化且单位方差变量) 是

$$\frac{\sum_{i=1}^{n} X_i - n\mu}{\sigma\sqrt{n}} = \frac{\overline{X}_n - \mu}{\frac{\sigma}{\sqrt{n}}} \tag{7.11}$$

CLT 的意思是：这个标准化变量的极限分布是 standard normal.

8.7.1 Berry-Esseen Theorem

Theorem Berry-Esseen, 收敛速度

设 $X_1, X_2, \cdots$ i.i.d., 且

$$\mathbb{E}[X_1] = \mu, \quad \mathbf{Var}[X_1] = \sigma^2 > 0, \quad \rho = \mathbb{E}\left[|X_1 - \mu|^3\right] < \infty \tag{7.12}$$

如果

$$Z_n = \frac{\overline{X}_n - \mu}{\frac{\sigma}{\sqrt{n}}}, \tag{7.13}$$

则存在 absolute constant (绝对常数) C, 使得

$$\sup_{z\in\mathbb{R}} |\Pr(Z_n \leq z) - \Phi(z)| \leq C\frac{\rho}{\sigma^3\sqrt{n}} \tag{7.14}$$

其中 Φ 是 $N(0,1)$ 的 CDF.

Note

CLT 只给出 asymptotic convergence (渐近收敛), Berry-Esseen theorem 给出误差速度。 在 third absolute moment (三阶绝对矩) 有限时，典型误差阶是 $\frac{1}{\sqrt{n}}$.